

Modelos de energía oscura dinámica
(Dynamical dark energy models)

Trabajo de Fin de Máster
para acceder al

**MÁSTER EN FÍSICA DE PARTÍCULAS Y DEL
COSMOS**

Autor: Jorge Martín Álvarez

Director: Airam Eduardo Marcos Caballero

Julio de 2026

Gracias a Airam por tratar el tema de este trabajo siempre con ilusión y con el máximo detalle, y por no haber dejado de apoyarme a pesar de mi progresión muy poco lineal.

A mis compañeros particulares y cosmólogos, por hacer de este año raro un tiempo tan llevadero y divertido.

A mis padres.

A Justine, a Marylin, a Jimena...

Abstract

Recent dark energy observations have created tension with the cosmological constant paradigm of the Λ CDM model, especially after the release of DESI DR2 in 2025. New data suggest evidence of a dynamical dark energy equation of state, whose parameter would sit at $w > -1$ at the present time and likely crossed the phantom divide $w = -1$ in the past, exhibiting a so-called *quintom* behaviour. In this work, we study models of dynamical dark energy that aim to reproduce the observed dependence of w with the scale factor. Our focus is put on field theories able to overcome the No-Go Theorem, which prevents too simple models from crossing the phantom divide smoothly. Concretely, we review the details of the h-essence model, and develop a generalisation thereof by replacing the quadratic term ϕ^2 in the h-essence Lagrangian with an arbitrary function $f(\phi)$. Using $f(\phi) = \tanh(\phi)$, we simulate the evolution of the dark energy density and equation of state for several sets of parameters, with the goal of producing compatible results with the observational data and previously published statistical reconstructions. We find three specific instances of our model which generate consistent evolution curves for at least one of the observables up to redshifts between $z = 0.3$ and $z = 1.5$, improving the results of the h-essence model.

Key words: dynamical dark energy, equation of state, phantom divide, quintom models.

Resumen

Observaciones recientes de energía oscura han dado lugar a una tensión con el paradigma de constante cosmológica del modelo Λ CDM, en especial tras la publicación de DESI DR2 en 2025. Los nuevos datos aportan evidencia a favor de una ecuación de estado dinámica de la energía oscura, cuyo parámetro se situaría en $w > -1$ en el momento actual y probablemente cruzó la línea *phantom* $w = -1$ en el pasado, mostrando un comportamiento conocido como *quintom*. En este trabajo, estudiamos modelos de energía oscura dinámica que tratan de reproducir la dependencia observada de w con el factor de escala. Nuestra atención se centra en teorías de campos capaces de evadir el Teorema No-Go, que impide cruzar la línea *phantom* a modelos muy simples. En concreto, repasamos los detalles del modelo de h-esencia, y desarrollamos una generalización de éste que consiste en sustituir el término cuadrático ϕ^2 del Lagrangiano de h-esencia por una función arbitraria $f(\phi)$. Empleando $f(\phi) = \tanh(\phi)$, simulamos la evolución de la ecuación de estado y la densidad de energía oscura para varios conjuntos de parámetros, con el objetivo de producir resultados compatibles con las observaciones y las reconstrucciones estadísticas publicadas en trabajos anteriores. Encontramos tres instancias específicas de nuestro modelo que dan lugar a una evolución compatible de al menos uno de los observables hasta redshifts entre $z = 0.3$ y $z = 1.5$, mejorando los resultados del modelo de h-esencia.

Palabras clave: energía oscura dinámica, ecuación de estado, línea *phantom*, modelos *quintom*.

Contents

1	Introduction	7
1.1	A brief history of dark energy	9
1.2	Recent observations	12
2	Review of existing dynamical models	17
2.1	Field models of dark energy	17
2.2	Models of Quintom Dark Energy	20
2.2.1	Multi-field models	21
2.2.2	Single field with higher derivatives	22
2.2.3	Modified gravity models	24
2.2.4	Effective field theory	25
2.2.5	Interacting dark energy	25
2.3	The h-essence model	26
2.3.1	Dynamics of h-essence	28
2.3.2	Numerical results	30
3	A new model: s-essence	34
3.1	Dynamics of s-essence	35
3.2	Numerical integration	37
3.3	Results	38
3.4	Discussion	44
4	Conclusions	46
	Bibliography	49

1 Introduction

As of 2026, the nature of dark energy remains one of the deepest mysteries in cosmology. Proposed by Einstein to avoid the collapse of a static universe subject to his field equations, we now regard dark energy as a physical agent that stretches the universe and favours the creation of spacetime, leading to the accelerated expansion that we observe. Following the 1998 observations that confirmed the accelerated expansion and thus the existence of dark energy, the concordance Λ CDM model emerged as a successful theory able to incorporate dark energy, in the form of a cosmological constant, while being consistent with observational data. However, the descriptions of dark energy in this and other models are purely effective, and the fundamental nature of this entity remain totally unknown.

To make things more complicated, recent observations seem to point toward the existence of *dynamical* dark energy. In other words, according to the more and more accurate cosmological data acquired by modern instruments, the average density of dark energy would not behave as a cosmological constant, but evolve with time just like the energy densities of matter and radiation do. These observations have led to the development of models which extend Λ CDM and produce dynamical dark energy through different mechanisms, aiming to reproduce observational data. Examples of such mechanisms include modified gravity, interacting dark energy and, more generally, effective field theories.

The goal of this thesis is to review the most relevant theories of dynamical dark energy and propose a new model that extends an existing effective field theory with the aim of improving its results. Specifically, we introduce *s-essence* as a generalisation of the theory of h-essence, by replacing the ϕ^2 factor in the h-essence Lagrangian with a general function of ϕ that gives rise to a whole family of models with the same structure.

Roughly speaking, recent observational data suggest that dark energy today behaves in a manner closer to matter, in the so-called *quintessence* regime; whereas not long in the past, it might have behaved as a stronger stretching force, known as the *phantom* regime. The line separating the two regions is known as the cosmological constant boundary, or phantom divide. Models that show both quintessence and phantom behaviour at different epochs are known as *quintom* models. According to the latest acquired data, this kind of models is the main candidate to describe dynamical dark energy, and therefore our work focuses on quintom models and the properties they are required to have. In particular, the theory of s-essence that we introduce is an example of a quintom model which crosses the phantom divide and produces consistent evolution curves for the dark energy equation of state and the dark energy density.

Taking this into account, we will dedicate the first part of the work to introduce quintom dark energy and a variety of models aiming to describe it,

and then present the model of s-essence that we developed. More concretely, the thesis is divided into three chapters, plus a final conclusions chapter.

In Chapter 1, we start by reviewing the basics of dark energy and its role in the concordance cosmological model and the Friedmann equations. Then, we continue by discussing alternative forms of dark energy beyond the cosmological constant. In particular, we introduce the w_0w_a CDM model and describe the quintessence and phantom regimes in terms of the equation of state of dark energy, which lead to quintom behaviour when combined. Finally, we review some of the most relevant observational data acquired during the last 25 years, notably the latest DESI release giving strong evidence of the existence of dynamical dark energy.

Chapter 2 is dedicated to exploring existing models of dynamical dark energy, with a focus on field-theoretic models. First, we derive the quintessence field model, in which this form of dark energy emerges from the existence of a real, scalar field. This is often used as a model for the cosmological constant itself, in the appropriate limit. Next, we state a No-Go theorem which prevents simple models from crossing the phantom divide. Hence, we then review modifications of the quintessence model that manage to cross the divide, and eventually focus on the h-essence theory. We derive this model in detail and present some basic results obtained with it.

Then, in Chapter 3, we generalise the idea of h-essence to create a new two-field model, that we name s-essence. After deriving its dynamics, we move on to numerically solving the equations of motion and compute the evolution of the main cosmological observables, namely the dark energy density and equation of state, for multiple sets of parameters. Our results can be considered successful, as s-essence provides solutions that cross the phantom divide smoothly and, more importantly, reproduce the observables reasonably well within the uncertainty margins available.

Lastly, in Chapter 4, we collect the conclusions of the work and discuss how the s-essence model might be modified to obtain more accurate results and exhibit some desirable features that it lacks in the presented form.

Notation and conventions

We use natural units and set $c = 1$ and $8\pi G = 1$ in all our calculations. We adopt the common convention in cosmology for the signature of the space time metric, that is $(-, +, +, +)$. Although most derivations are carried out for a general metric $g^{\mu\nu}$, in specific applications we always use the FLRW metric. When working with indices, we use the Einstein summation convention. Furthermore, we will repeatedly use the following abbreviation:

$$(\partial\phi)^2 = g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi, \quad (1.1)$$

where $g^{\mu\nu}$ is a metric tensor and ϕ is a field.

Over the work, cosmic time is denoted by t , but we will mainly use the scale factor a and the redshift z as time indicators. Recall that these variables are related by $1 + z = 1/a$.

1.1 A brief history of dark energy

Dark energy, in the form of a cosmological constant, was first conceived by A. Einstein in 1917 as the mechanism to prevent the gravitational collapse predicted by General Relativity and hold the static universe that was believed to exist in that time. However, the discovery over the following decade that the universe was expanding removed the need for a cosmological constant, as the expansion itself served to counteract the attracting effect of gravity. The latter would only slow down the rate of expansion and eventually stop it, providing a consistent explanation for the observations without an exotic dark energy component.

In 1998, though, cosmological measurements of distant type Ia supernovae dramatically changed this vision. Comparison of the luminosity distance and redshift of these objects revealed that the expansion of the universe is accelerating, and thus Einstein's introduction of a cosmological constant was actually correct! Later observations of the cosmic microwave background (CMB) and the large-scale structure of the universe confirmed the accelerated expansion. The proposed physical agent for this phenomenon is *dark energy*: a species with negative pressure which therefore favours the creation of vacuum and would dominate today's universe at large scales [4].

Together with dark energy, evidence of an invisible, massive component in galaxies and galaxy clusters, named *dark matter*, gave rise to the concordance Λ CDM cosmological model. This model is based on the equations of General Relativity and features four main energy components in the universe: radiation, ordinary ("baryonic") matter, dark matter and dark energy. Unlike all other species, whose density decays as the scale of the universe grows, the density of dark energy remains constant in the Λ CDM model, making it the dominant species at the present epoch as the accelerated expansion suggests.

The dynamics of a universe subject to the Λ CDM model are governed by the Friedmann equations, which can be derived from Einstein's field equations. The general form of the Friedmann equations is the following:

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{1}{3}\rho - \frac{\kappa^2}{a^2} + \frac{\Lambda}{3}, \quad (1.2)$$

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p) + \frac{\Lambda}{3}. \quad (1.3)$$

Here, a is the scale factor of the universe, set at $a_0 = a(t_0) = 1$ at the present time; H is the Hubble parameter; $\rho = \rho_r + \rho_m$ is the total energy density of matter and radiation; κ is a constant indicating the curvature of the universe; and Λ is the cosmological constant.

Observations tend to indicate that the universe is approximately flat ($\kappa \simeq 0$), so we will neglect this term over the whole work. On the other hand, it is common to absorb the cosmological constant into the total density, defining the dark energy density as $\rho_\Lambda = \Lambda$. Then, Eqs. (1.2) and (1.3) become, respectively

$$H^2 = \frac{1}{3}\rho_{\text{tot}}, \quad (1.4)$$

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p - 2\rho_\Lambda). \quad (1.5)$$

Here $\rho_{\text{tot}} = \rho + \rho_\Lambda = \rho_r + \rho_m + \rho_\Lambda$ is the total energy density taking also dark energy into account.

Now, if we assume dark energy behaves as a perfect fluid, the second Friedmann equation (Eq. (1.5)) should include a term $\rho_\Lambda + 3p_\Lambda$, as for the other species. This leads to equating

$$-2\rho_\Lambda = \rho_\Lambda + 3p_\Lambda, \quad (1.6)$$

which yields $p_\Lambda = -\rho_\Lambda$. Recall that, for a fluid, the relation $p = p(\rho)$ is called the *equation of state*. In cosmology, it is common that this equation is linear, of the form $p = w\rho$. In an abuse of terminology, the parameter w itself is often called equation of state. Thus, dark energy in the form of a cosmological constant Λ presents an equation of state $w = -1$.

It is worth noting that, more generally, if we were to consider a different model than Λ CDM in which dark energy has a constant equation of state w_{DE} , the accelerated expansion of the universe requires $w_{\text{DE}} < -1/3$. Indeed, for both matter and radiation $w \geq 0$, and so $\ddot{a} \leq -Ca$ for some $C \geq 0$, leading to a decelerating universe. In turn, the contribution of dark energy is given by

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho_{\text{DE}} + 3p_{\text{DE}}) = -\frac{1}{6}\rho_{\text{DE}}(1 + 3w_{\text{DE}}), \quad (1.7)$$

and then $\ddot{a} > 0$ only if $w_{\text{DE}} < -1/3$. This observation will become important when discussing models of dynamical dark energy in Chapters 2 and 3.

Variations of the Λ CDM model

As we will review in the following section, recent cosmological observations suggest the possibility that the dark energy equation of state deviates from the expression $w = -1$ from the concordance Λ CDM models. Subsequently, alternative models that introduce small modifications are considered. Due to the lack of a fundamental theory able to explain dark energy from first principles, such modifications consist in simple generalisations aiming to fit a small number of parameters using the available data. The most basic is known as the w CDM model, in which the cosmological constant equation of state $w = -1$ is replaced by an arbitrary w still constant in time. As reasoned

before, this parameter is already constrained to remain below $-1/3$ by the very nature of dark energy.

A more general model is w_0w_a CDM, in which the time evolution of w depends on two parameters. It is common to use the Chevallier-Polarski-Linder (CPL) parametrization, introduced in [5] and [19], which relates w with the scale factor a as

$$w(a) = w_0 + w_a(1 - a). \quad (1.8)$$

Note that this amounts to taking a linear approximation of the function $w(a)$ at $a = 1$. Hence, if we choose w_0 and w_a within a suitable region of the parameter space, the equation of state may transition from $w < -1$ to $w > -1$ or vice versa at some scale factor $a > 0$. This is a very interesting setting, since the cosmological constant $w = -1$ marks a boundary between two very different regimes.

Models with (constant) $w > -1$ are known as **quintessence** dark energy models. The behaviour of the universe in these models is not too different from the Λ CDM universe: solving the Friedmann equations for an arbitrary w leads to the relation

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{-3(w+1)}, \quad (1.9)$$

where $w + 1 > 0$ in this case. For this reason, the density of quintessence dark energy decreases as the universe expands, but the condition $w < -1/3$ ensures that the decrease is slower than a^{-2} , ensuring that dark energy is the dominating species at large times. Summarizing, we can regard quintessence as a form of dark energy which behaves in an intermediate way between matter and the cosmological constant, but whose effects do not differ much from those of the latter.

In turn, models with (constant) $w < -1$ are known as **phantom** dark energy models. For these models, satisfying $w + 1 < 0$, the density of dark energy increases with the scale factor! This means not only that energy is created from the vacuum as spacetime grows, but that a larger amount of energy is created the larger spacetime is. Furthermore, phantom dark energy has the additional issue that such a universe would violate the *null-energy condition* of the stress-energy tensor $T_{\mu\nu}$, stating that any null-vector n^μ must satisfy the relation

$$T_{\mu\nu}n^\mu n^\nu \geq 0. \quad (1.10)$$

Otherwise, non-desirable consequences like the breakdown of causality or the violation of the second law of thermodynamics arise [1]. As we will show later, when modelling phantom dark energy as emerging from a field, the problem becomes clearer due to the presence of a negative kinetic term that leads to an energy profile with no minimum! Because of this, phantom dark energy is generally regarded as an unphysical solution. Nevertheless, a

common perspective among cosmologists is that phantom models should still be probed as long as experiments do not rule this regime out, also serving as a test for alternative gravity theories which predict this behaviour of dark energy [17].

In addition, as discussed in [27], the null-energy condition may be satisfied if dark energy stays in the phantom regime for a short amount of time. This behaviour is already found in other physical phenomena like the Casimir effect or Hawking evaporation [1]. In our case, it may be achieved by considering a second order expansion of $w(a)$, namely

$$w(a) = w_0 + w_a \left(1 - a + \frac{3}{16} a^2 \right). \quad (1.11)$$

Notice that now, $w(a)$ has one minimum (provided $w_a > 0$), and is allowed to cross the cosmological constant boundary twice while staying only shortly in the phantom regime. This parametrization is rarely used and, more importantly, observations seem to favour a negative w_a , as we show in the next section. However, it still provides an example of a model able to overcome the phantom problem.

A model which transits from one regime to the other is generally known as a **quintom** model. In the linear model with only one transition, we speak of quintom-A if the transition takes place from quintessence to phantom, and of quintom-B if the transition happens from phantom to quintessence.

Quintom dark energy poses a more intricate problem than the null-energy condition of phantom energy. In short, models crossing the cosmological constant boundary, also known as the *phantom divide*, are likely to give rise to divergences at perturbation level [17]. Actually, there is a No-Go Theorem [33] specifying a class of models for which crossing the phantom divide is strictly forbidden. Unfortunately, this class contains the most fundamental dark energy models, including the basic quintessence and phantom field theories. Thus, the focus needs to be set on finding quintom models out of that class that cross the phantom divide smoothly. We review the exact formulation of the theorem and discuss examples of allowed quintom models in Section 2.2.

1.2 Recent observations

In recent years, improvements in the precision of cosmological observations have produced more and more accurate results about the nature of dark energy, and in particular of the value of the equation of state w_{DE} . Following the complete survey in [4], we review some of these measurements up to the recent DESI DR2 observations, which suggest strong evidence of a dynamical behaviour of dark energy. A summary of the data published in different studies is collected in Table 1.

The first precise measurements of the equation of state w were done by WMAP between 2003 and 2012 [30, 31, 15, 16, 2]. These results were compatible with a cosmological constant, while constraining the range of values for w around -1 . In 2013, the first results from Planck [25] seemed to favour a phantom dark energy scenario with $w = -1.13^{+0.13}_{-0.14}$ (Planck + WMAP9 + SLNS). However, later improvements in data calibration led to a far less significant result $w = -1.006^{+0.085}_{-0.091}$ [26], which gave no evidence to reject the cosmological constant paradigm as of 2015.

Study	w	w_0	w_a
WMAP1 [30]	-0.98 ± 0.12	-	-
WMAP3 [31]	$-0.967^{+0.073}_{-0.072}$	-	-
WMAP5 [15]	$-1^{+0.12}_{-0.14}$	-1.06 ± 0.14	0.36 ± 0.62
WMAP7 [16]	-1.10 ± 0.14	-0.93 ± 0.13	$-0.41^{+0.72}_{-0.71}$
WMAP9 [2]	$-1.073^{+0.090}_{-0.089}$	-	-
Planck2013 [25]	$-1.13^{+0.13}_{-0.14}$	-	-
Planck2015 [26]	$-1.006^{+0.085}_{-0.091}$	-	-
Pantheon+(SN) [3]	-0.90 ± 0.14	-	-
Pantheon+ [3]	$-0.978^{+0.024}_{-0.031}$	-	-
Union3 [29]	-	$> -1 \ (1.7 - 2.6\sigma)$	$< 0 \ (1.7 - 2.6\sigma)$
DESY5 [7]	$> -1 \ (1\sigma)$	-	-
DESI DR1 [8]	-	$> -1 \ (2.6 - 3.9\sigma)$	$< 0 \ (2.6 - 3.9\sigma)$
DESI DR2 [9]	-	$> -1 \ (2.8 - 4.2\sigma)$	$< 0 \ (2.8 - 4.2\sigma)$

Table 1: Summary of the estimates of w (Λ CDM model) or w_0 and w_a (w_0w_a CDM model) obtained by several studies on dynamical dark energy.

The situation started to change with the new observational data sets obtained by DESY1, Pantheon, Planck2018 and DESY3 after 2018. The estimations of w remained consistent with $w = -1$, namely $w = -0.90 \pm 0.14$ (SN data only) and $w = -0.978^{+0.024}_{-0.031}$ (SN, CMB and BAO) [3]. Nonetheless, the values of w_0 and w_a obtained when fitting the CPL models presented slight deviations from Λ CDM, favouring values $w_0 > -1$ and $w_a < 0$ with a significance of $1.7 - 2.6\sigma$ [29]. In 2024, new data obtained by DESY5 [7] showed evidence of $w > -1$ even when fitting w alone, with significance of more than 1σ and for multiple combinations of data, such as supernovae alone or combined with CMB, BAO and 3×2 pt measurements. The BAO measurements released by DESI (DR1) [8] also in 2024 for the w_0w_a CDM models are in agreement with these results, increasing the significance to 2.5σ and higher in different combinations with previous data.

Finally, DESI released a second set of data (DR2) [9] in 2025, which reduced the uncertainties of DR1. In this case, the Λ CDM model is rejected with a significance of 3.1σ when using the BAO data alone, and between 2.8σ and 4.2σ when combining them with other data sources. The posterior

distributions of w_0 and w_a emerging from these combinations of data are shown in Fig. 1. Again, the data favour more clearly a combination $w_0 > -1$ and $w_a < 0$, with $w_a + w_0 < -1$. In other words, in the first order approximation, observations suggest the existence of quintom-B dark energy, with an equation of state w that grows with the scale factor and crossed the phantom divide at some point in the past, since the condition $w_a + w_0 < -1$ implies $w < -1$ at the Big Bang ($a = 0$). Observe that the grey-shaded areas of Fig. 1 represent the regions of the parameter space leading to pure phantom and pure quintessence behaviour, which do not intersect even the 95% posterior distributions of any combination of data.

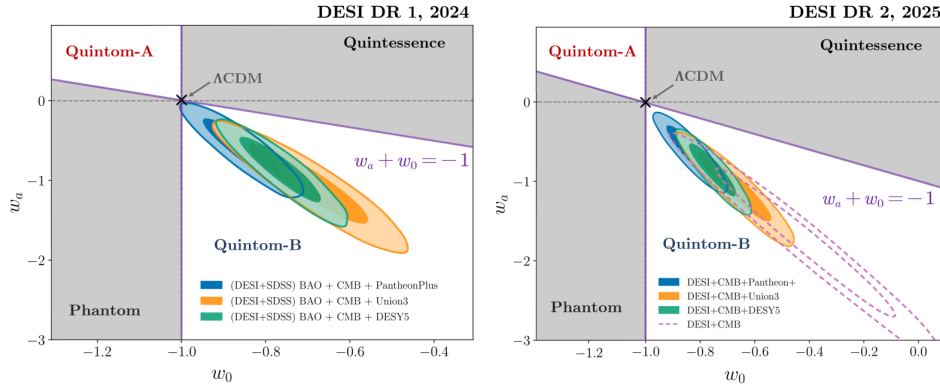
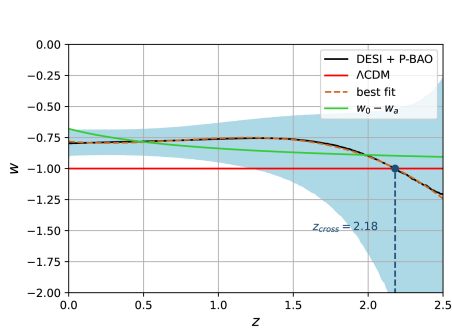


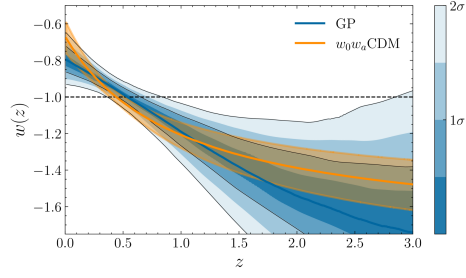
Figure 1: 68% and 95% posterior distributions for the parameters w_0, w_a of the $w_0 w_a$ CDM model, obtained from the combination of the DESI BAO data with CMB and supernovae data from Pantheon+, Union3 and DESY5. Figures were originally published in [8, Fig. 6] (left) and [9, Fig. 11] (right). The presented layout is taken from [4, Fig. 1].

Nonetheless, any linear approximation of the function $w(a)$, such as the CPL parametrization, can only be effective at low redshifts. Thus, in order to find evidence for crossing of the phantom divide, non-parametric approaches are also necessary. In that regard, Gaussian process regression is a widely used technique, which relies on a certain kernel covariance function to reconstruct a function and its derivatives from a given set of data points.

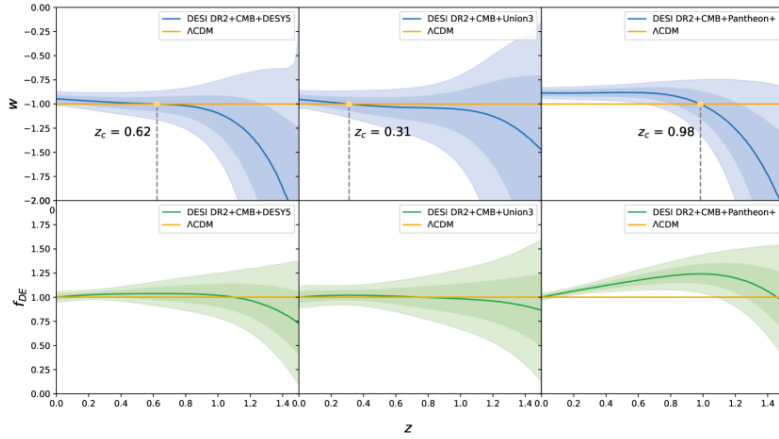
Several studies [38, 40, 10] have employed Gaussian process regression with different datasets to reconstruct the function $w(a)$, or equivalently $w(z)$ as this is the most common parametrisation when using the reconstruction method. Fig. 2 depicts the results obtained. All three studies considered agree on the quintom-B behaviour of dark energy, with crossing of the phantom divide happening between $z = 0.31$ and $z = 2.18$. However, it is interesting to notice that the shape of the function $w(a)$ significantly differs among the studies and the datasets used. In [10] (Fig. 2a), and [40] with combined data from DESI DR2, CMB and DESY5 or Union3 (left and



(a) DESI+BAO [38]. The shaded area represents the 68% confidence interval.



(b) DESI+CMB+Union3 [10]. The blue-shaded areas represent 1σ and 2σ uncertainty regions. The orange-shaded area represents the 1σ uncertainty region for the w_0w_a CDM fit.



(c) DESI+CMB+(DESY5, Union3 or Pantheon+) [40]. The shaded areas indicate 1σ and 2σ uncertainty regions.

Figure 2: Results of the studies that employed Gaussian process regression to reconstruct the function $w(z)$ using different combinations of data. The solid curve represents the mean value, with uncertainties indicated by the shaded areas in each case. The top panels indicate as well the best fit of the w_0w_a CDM model, using the CPL parametrisation.

central panels of Fig. 2c), w evolves very close to the CPL approximation until crossing the phantom divide. In contrast, in [38] (Fig. 2b) and [40] with data from DESI DR2, CMB and Pantheon+ (right panel of Fig. 2c), w describes a plateau at low redshift and even grows slightly with z before reaching a tipping point and falling to cross the phantom divide.

Interestingly, [40] also shows the evolution of the energy density of dark energy with each combination of datasets (Fig. 2c). Whereas the reconstructions using DESY5 and Union3 leave ρ_{DE} essentially constant until well after crossing the phantom divide, the one using Pantheon+ instead leads to ρ_{DE} increasing with z with a maximum at $z \simeq 1$. This is consistent with the later crossing of the phantom divide with the Pantheon+ data, since $w = p/\rho$ and thus a less negative w corresponds to a larger energy density. A similar effect, although not so marked, is observed in [10].

In any case, this kind of reconstructions can, again, be trusted only locally. Despite giving more information than first-order approximations, since they allow to obtain a varied range of shapes for the reconstructed function, the confidence intervals become extremely large around $z = 2$, entailing huge uncertainty about the real shape of the function in that region.

Summarising, the statistical analysis of cosmological data gathered during the last 25 years suggests that dark energy does not behave as a cosmological constant, but rather presents a dynamic equation of state $w(a)$. At the present time, there is evidence for $w_0 > -1$, in opposition to the Λ CDM value $w_\Lambda = -1$. Furthermore, both the estimate of the parameter w_a from the CPL parametrization and the reconstructions using Gaussian process regression favour a growing behaviour of the function $w(a)$. In turn, this points to a crossing of the phantom divide at some redshift between $z = 0.3$ and $z = 2$.

The confirmation of such a tension between recent cosmological observations and the concordance Λ CDM model has created a renewed interest in theoretical models able to describe dark energy and its equation of state as a dynamical function of time, or the scale factor. These models should aim to replicate the local behaviour of the observables discussed above, and then propose consistent evolutions for further redshifts. We dedicate the next chapter to reviewing the main types of models that give rise to dynamical dark energy.

2 Review of existing dynamical models

Given the observational data discussed in the previous chapter, different approaches attempting to model dynamical dark energy have emerged in recent years. In this chapter, we present a series of such models, focusing our interest on quintom dark energy models.

For that, we start with the most basic field model of quintessence dark energy, consisting of a single, real, scalar field evolving under the influence of a certain potential. We also review its phantom counterpart. Then, we move on to stating the No-Go Theorem, which gives constraints on the form of a model able to cross the phantom divide, and list the main extensions of quintessence that manage to produce quintom dark energy. At the end of the chapter, we derive the model of h-essence, which is the basis of the s-essence model that we introduce in Chapter 3.

2.1 Field models of dark energy

Field-theoretic approaches have proved to be very successful through the history of physics, providing a common framework for an effective description of forces and particles, from classical electromagnetism to the quantum physics of the Standard Model. Here, we present one of the simplest field models for (dynamical) dark energy. Strictly speaking, the model describes quintessence dark energy, as it gives rise to an equation of state $w > -1$. However, in the appropriate limit, the same model can be used to describe a cosmological constant dark energy with $w \gtrsim 1$. Similarly, a sign twist will allow us to model phantom dark energy within the same formalism. In fact, the multi-field approach that we introduce in Section 2.2.1 is no more than the natural extension of this fundamental model to an arbitrary number of fields.

Thus, let us assume that dark energy emerges from a scalar field $\phi(\mathbf{x}, t)$ defined on the whole spacetime, known as the quintessence field. We assume that the field has an associated Lagrangian density

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - V(\phi). \quad (2.1)$$

Here, $V(\phi)$ is a real-valued potential. A common choice is to set

$$V(\phi) = \lambda \phi^\alpha \quad (2.2)$$

where α and λ are real constants and $\lambda > 0$. In turn, for the kinetic term, we use the abbreviated notation

$$(\partial\phi)^2 := g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \equiv \partial^\mu \phi \partial_\mu \phi, \quad (2.3)$$

where we adopt the sign convention $(-, +, +, +)$ for the metric as in the rest of the work. Note that the uncommon negative sign of this term in Eq. (2.1)

is only apparent, since the time derivative carries an additional minus sign that brings back the positive sign to the kinetic energy. For this reason, we will henceforth refer to this Lagrangian as having a *positive* kinetic term, counter-intuitive as it may seem.

In fact, assuming homogeneity, the spatial derivatives of ϕ vanish and Eq. (2.3) becomes simply

$$(\partial\phi)^2 = -\dot{\phi}^2, \quad (2.4)$$

where $\dot{\phi} := \partial_t \phi$, the derivative with respect to cosmic time. With that, the Lagrangian reduces to

$$\mathcal{L} = \frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (2.5)$$

In this and the next chapter, we will carry out the derivations in whole generality, that is, using the full expression for $(\partial\phi)^2$. However, we will often assume homogeneity to simplify the equations that we derive. On the one hand, the majority of cosmological studies are done under this assumption, since there is no observational evidence of the contrary at large scales. On the other hand, this will allow us to obtain final expressions that are easier to interpret and to deal with numerically, which are desirable properties of a toy model like the one we will develop here.

The next step is to relate the field theory described by Eq. (2.5) to actual cosmological observables, namely pressure and energy density of dark energy, and with those the equation of state.

Assuming dark energy behaves as a perfect fluid, and once more in a homogeneous setting, we can write the stress-energy tensor as (see [22, p. 132])

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu}, \quad (2.6)$$

where, working in the fluid's proper frame, we may take $u^\mu = (1, 0, 0, 0)$. The stress-energy tensor is related to the Lagrangian density by (see [24, p. 19])

$$T_\nu^\mu = -\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\nu\phi + \delta_\nu^\mu\mathcal{L}, \quad (2.7)$$

where δ_ν^μ is the Kronecker delta, i.e. $\delta_\nu^\mu = 1$ if $\mu = \nu$ and 0 otherwise. Note that, compared to the source, we use opposite signs here. This is because we use the cosmology definition of the stress-energy tensor, which normally carries a negative sign in quantum field theory, and thus the relation with the Lagrangian also has opposite signs.

In order to express density and pressure in terms of the Lagrangian, we will relate them to the diagonal entries of the stress-energy tensor with mixed indices. Observe that

$$T_\nu^\mu = g_{\nu\alpha}T^{\mu\alpha} = g_{\nu\alpha}(\rho + p)u^\mu u^\alpha + pg_{\nu\alpha}g^{\mu\alpha} = g_{\nu\alpha}(\rho + p)u^\mu u^\alpha + p\mathbb{1}_\nu^\mu, \quad (2.8)$$

where $\mathbb{1}$ represents the identity tensor. Hence,

$$T_0^0 = g_{0\alpha}(\rho + p)u^0u^\alpha + p = -(\rho + p) + p = -\rho, \quad (2.9)$$

$$T_i^i = g_{i\alpha}(\rho + p)u^iu^\alpha + p = 0 + p = p. \quad (2.10)$$

Finally, from Eq. (2.7) we can calculate:

$$\rho = -T_0^0 = -\left(-\dot{\phi}^2 + \mathcal{L}\right) = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad (2.11)$$

$$p = T_i^i = -0 + \mathcal{L} = \frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (2.12)$$

With that, we may write the equation of state of the quintessence field model as

$$w = \frac{p}{\rho} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} = \frac{\dot{\phi}^2 - 2V(\phi)}{\dot{\phi}^2 + 2V(\phi)}. \quad (2.13)$$

In order to evaluate the behaviour of w , let us start by assuming $\dot{\phi}^2 < 2V(\phi)$, which we may dub a “not-too-fast rolling” condition. From Eq. (2.2), the potential term is always positive, and so this implies

$$-1 < \frac{-2V(\phi)}{\dot{\phi}^2 + 2V(\phi)} < w < 0. \quad (2.14)$$

With a slightly more constraining inequality, we can improve the upper bound to $w < -1/3$, and then the field ϕ sits in the correct range $(-1, -1/3)$ to describe quintessence dark energy.

We can be even more restrictive and impose a slow rolling condition, that is $\dot{\phi}^2 \ll V(\phi)$, mimicking the approximation used with the inflation field. Then, the equation of state evolves as

$$w = \frac{\varepsilon - 1}{\varepsilon + 1} \approx -1 + 2\varepsilon \quad (2.15)$$

with $\varepsilon \ll 1$. Hence, the nature of dark energy in this regime is essentially that of a cosmological constant.

Finally, switching the sign of the kinetic term in Eq. (2.1) would give rise to phantom dark energy. Now, the general expression for the Lagrangian is

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - V(\phi), \quad (2.16)$$

and we will say that it features a *negative* kinetic term, as it leads to negative kinetic energy when expanding the square with the metric. Subsequently, if ϕ is homogeneous, we have

$$\mathcal{L} = -\frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (2.17)$$

This sign change is propagated to Eqs. (2.11) and (2.12), yielding an equation of state

$$w = \frac{p}{\rho} = \frac{-\dot{\phi}^2 - 2V(\phi)}{-\dot{\phi}^2 + 2V(\phi)}. \quad (2.18)$$

Then, in the not-too-fast rolling condition, w is bounded *above* by -1 , realising the nature of phantom dark energy. In the slow rolling limit, this model also approaches the cosmological constant, but from below.

Of course, the negative kinetic term gives rise to the null-energy condition and related problems discussed when introducing phantom dark energy in Chapter 1. In particular, in such a model there is no fundamental state, since for a fixed potential $V(\phi)$, the energy density function

$$\rho = -\frac{1}{2}\dot{\phi}^2 + V(\phi) \quad (2.19)$$

has no minimum. Hence, a single-field theory of phantom dark energy is physically unfeasible. Nonetheless, a mixed solution in the form of quintom dark energy does leave room for models that transit through the phantom regime while effectively reproducing physical observables. We study this kind of models in the next section.

2.2 Models of Quintom Dark Energy

The recent cosmological observations discussed in Section 1.2 point to a dynamical nature of dark energy, which likely crossed the phantom divide at some point in the past. The basic field models for quintessence and phantom reviewed at the beginning of the chapter are not able to describe such behaviour, as the equation of state is bounded either above or below by the cosmological constant $w = -1$. Hence, it is necessary to build alternative models which do manage to cross the cosmological constant boundary, or phantom divide.

Nevertheless, as we anticipated in Chapter 1, the nature of these models is restricted by the No-Go Theorem that we state next [33].

Theorem (No-Go Theorem). *Consider a theory of dark energy in a FLRW universe described by a single scalar field ϕ with a Lagrangian $\mathcal{L}(\phi, (\partial\phi)^2)$, minimally coupled to Einstein gravity. Then, the equation of state w of dark energy cannot cross the cosmological constant boundary, $w = -1$.*

In other words, single-field models like quintessence or phantom are too simple to cross the phantom divide. Successful models need to have more degrees of freedom, by featuring either two or more fields, a more complex Lagrangian, further couplings with other species or forces, or a different gravity theory.

The current section is devoted to introducing each of these possible modifications. We use the survey [4] as the main reference and partially follow the outline and derivations presented there.

We put a special focus on multi-field models (Section 2.2.1), since the model we develop in Chapter 3 is of this kind. In particular, here we establish a general framework for multi-field models in terms of a metric tensor defined on the field space that determines how each pair of fields contributes to the Lagrangian. Such a structure is already present in the most basic two-field model and can be extended naturally to an arbitrary number of fields. In addition, the introduction of this formalism will pave the way for studying the h-essence model in Section 2.3, which serves as the benchmark for the more general version in Chapter 3.

Then, in Section 2.2.2, we review the details of a single-field model with higher derivatives in the Lagrangian, given that the methodology used is similar to ours and, like h-essence, it provides another early example of smooth crossing of the phantom divide. For the remaining theories, we give a brief description and supply references in which deeper studies of each can be consulted.

2.2.1 Multi-field models

At the beginning of the chapter, we saw the basic field model for quintessence, governed by the Lagrangian in Eq. (2.1). Here, dark energy was produced by a single, real field ϕ , with associated potential $V(\phi)$, in such a way that $w \gtrsim -1$ if ϕ varies slowly with time, namely $\dot{\phi}^2 \ll V(\phi)$.

The simplest generalisation of this model that leads to quintom dark energy consists in adding a second real field θ contributing with a *negative* kinetic term to the dark energy Lagrangian, as in

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}(\partial\theta)^2 - V(\phi, \theta). \quad (2.20)$$

In this way, when the term $(\partial\phi)^2$ dominates over $(\partial\theta)^2$, the Lagrangian reduces to Eq. (2.1), producing quintessence dark energy, which in the slow-roll limit behaves as a cosmological constant. Conversely, if $(\partial\theta)^2$ is the dominating term, the model behaves like Eq. (2.16), and the equation of state is located under the phantom divide, that is $w < -1$. Hence, if one manages to construct a model in which the relevant field switches from θ to ϕ , the nature of dark energy will transit from phantom to quintessence, mimicking the quintom-B evolution. Such a two-field model was first proposed in [12].

Notice that the kinetic part of Eq. (2.20) reminds of the expression for the line element in a two-dimensional hyperbolic space, where fields represent coordinates in each dimension. In other words, we can define a metric tensor γ^{jk} in the field space, given in this case by $\gamma^{jk} = \text{diag}(1, -1)$, and express

the Lagrangian in a more compact way as

$$\mathcal{L} = -\frac{1}{2}\gamma^{jk}(\partial\phi_j)(\partial\phi_k) - V(\Phi). \quad (2.21)$$

where $\phi_1 := \phi$ and $\phi_2 := \theta$ and $\Phi = (\phi_1, \phi_2)$. In general, the metric tensor γ^{jk} might depend on the fields ϕ_j .

This is an interesting framework for dark energy field theories. Indeed, we can generalise the one-field theories of quintessence and phantom dark energy using this formalism, taking the one-dimensional metrics $\gamma = 1$ and $\gamma = -1$, respectively. In turn, Eq. (2.21) is equally valid if we extend our model to an arbitrary number of fields $\phi_1, \dots, \phi_n$, by letting j, k run from 1 to n and $\Phi = (\phi_1, \dots, \phi_n)$. Within this context, an n -dimensional Euclidean metric defined by the matrix $\pm \text{diag}(1, \dots, 1)$ would represent an n -field quintessence or phantom model, depending on the sign. Matrices with a mixture of signs in the diagonal serve, instead, to build multi-field quintom models, generalising the two-field model from Eq. (2.20).

Concretely, the family of diagonal multi-field quintom models is given by matrices of the form $\text{diag}(\varepsilon_1, \dots, \varepsilon_n)$, where $\varepsilon_j \in \{-1, 1\}$ and the entries are ordered, i.e. $\varepsilon_j \geq \varepsilon_k$. Like this, we recover the Lagrangian density from [32],

$$\mathcal{L} = -\frac{1}{2} \sum_{j=1}^n \varepsilon_j (\partial\phi_j)^2 - V(\phi_1, \dots, \phi_n). \quad (2.22)$$

Assuming homogeneity, the pressure and energy density of a diagonal multi-field model are given by

$$\rho = \frac{1}{2} \sum_{j=1}^n \varepsilon_j \dot{\phi}_j^2 + V(\phi_1, \dots, \phi_n), \quad (2.23)$$

$$p = \frac{1}{2} \sum_{j=1}^n \varepsilon_j \dot{\phi}_j^2 - V(\phi_1, \dots, \phi_n). \quad (2.24)$$

Thus, the equation of state reads

$$w = \frac{\sum_j \varepsilon_j \dot{\phi}_j^2 - 2V(\phi_1, \dots, \phi_n)}{\sum_j \varepsilon_j \dot{\phi}_j^2 + 2V(\phi_1, \dots, \phi_n)}. \quad (2.25)$$

As we mentioned before, we will revisit multi-field models at the end of the chapter. Concretely, we will introduce the details of the h-essence model, which provides an example of successful crossing of the phantom divide and serves as the base for the model we develop in Chapter 3.

2.2.2 Single field with higher derivatives

A second modification of the quintessence Lagrangian consists in including higher derivatives in the kinetic term, namely a coupling term between $X := (\partial\phi)^2/2$ and the higher derivative $\square\phi := \nabla_\mu \nabla^\mu \phi$.

There are multiple forms for a Lagrangian featuring such a coupling. Here we will present the one in [18]. As an illustrative model able to cross the $w = -1$ boundary, the authors consider a system without potential, $V(\phi) \equiv 0$, and a Lagrangian given by

$$\mathcal{L} = -X + c_1 X \Box \phi + c_2 X \phi^2. \quad (2.26)$$

For the sake of comparison, observe that the single-field model with zero potential has $\mathcal{L} = -X$ using this notation.

From Eq. (2.7), one can obtain the density and pressure associated to $\mathcal{L}$,

$$\rho = \left(c_2 \phi^2 - 1 + 6c_1 H \dot{\phi} \right) X, \quad (2.27)$$

$$p = \left(c_2 \phi^2 - 1 - 2c_1 \ddot{\phi} \right) X. \quad (2.28)$$

Then, the equation of state reads

$$w = \frac{c_2 \phi^2 - 1 - 2c_1 \ddot{\phi}}{c_2 \phi^2 - 1 + 6c_1 H \dot{\phi}}. \quad (2.29)$$

In [18], the authors calculate the equation of motion and carry out a numerical integration of the system, for a fixed c_1 and two values of c_2 . They set initial conditions $\phi_i = -1.2$ and $\dot{\phi}_i \gtrsim 0$.

In the first case ($c_2 = 0$), the equation of state reduces to

$$w = \frac{-1 - 2c_1 \ddot{\phi}}{-1 + 6c_1 H \dot{\phi}}. \quad (2.30)$$

First, this equals essentially 1 (as $\dot{\phi}_i \approx \ddot{\phi}_i \approx 0$) and then is shown to approach -2 during matter-domination, crossing the phantom divide downwards. In turn, the expression tends to -1 in the limit $t \rightarrow \infty$. The numerical results agree with these limit calculations.

The second case ($c_2 > 0$) is even more interesting, as the phantom divide is crossed upwards, matching the most likely transition suggested by observations. Again, there is a first crossing from quintessence to phantom at early times, while the opposite crossing happens for a slightly above 1.

In [4], this 2012 model is compared with the newly released DESI data in a neighbourhood of $a = 1$. The confidence regions for w_0 and w_a allow to obtain areas in the parameter space of c_1 and c_2 within which the results of the higher-derivative model are compatible with DESI, and these areas turn out to be quite large. Hence, this kind of models is confirmed as a successful alternative able to cross the phantom divide near the present time.

There are similar models that modify the quintessence kinetic term using different functions of ϕ or its derivatives. Due to its notable historical

relevance as the earliest such solution, we briefly mention *k-essence* (from *kinetic* essence) [6]. This theory replaces the kinetic term X by a function

$$P(\phi, X) = K(\phi)X + L(\phi)X^2 + \dots, \quad (2.31)$$

where the freedom in the functions of ϕ that act as coefficients and the higher powers of X can overcome the No-Go Theorem. In [6], the authors show that such models can give rise to both quintessence and phantom dark energy. However, they highlight that the k-essence requires a delicate choice of initial conditions to produce phantom behaviour, and do not attempt to develop a quintom solution. In modern literature, this theory has not been proposed as a solution for the phantom divide crossing problem.

2.2.3 Modified gravity models

Both the multi-field and the higher-derivative models are built within the same flat FLRW universe relying on the celebrated general relativity theory. Instead, modified gravity theories propose a more fundamental way to overcome the phantom divide problem, by altering certain laws of physics and subsequently the evolution of dark energy in the universe.

In the context of dynamical dark energy, a commonly used formalism is that of metric-affine gravity [14, 13]. The idea of metric-affine gravity is to introduce a *general affine connection*, which transforms the Levi-Civita connection by adding a *disformation* and a *contortion* term. This change leads to new expressions for the main geometric quantities of spacetime, including the curvature, torsion and non-metricity tensors and the corresponding scalars R , T and Q [13]. Under certain conditions, the general action can be written in terms of only one of these quantities, giving rise to the theories of $f(R)$, $f(T)$ and $f(Q)$ gravity. Namely,

$$S = \int \left(\frac{1}{2}f(A) + \mathcal{L}_m \right) \sqrt{-g} d^4x, \quad (2.32)$$

where f is an arbitrary function and $A \in \{R, T, Q\}$.

In relation with dark energy, each of the three theories provides an expression for ρ_{DE} and p_{DE} in terms of the quantity A and the function $f(A)$. These are then plugged into the modified Friedmann equations

$$H^2 = \frac{1}{3}(\rho_m + \rho_{\text{DE}}), \quad (2.33)$$

$$-2\dot{H} - 3H^2 = p_m + p_{\text{DE}}. \quad (2.34)$$

The concrete study of these three theories goes beyond the scope of this work. Nonetheless, it is important to remark that specific forms of the function f have been found to produce quintom dark energy compatible with observations [28, 39], making modified gravity another strong candidate to describe dynamical dark energy.

2.2.4 Effective field theory

Effective field theory provides a more general framework for the study of dynamical dark energy, which actually contains one-field theories and modified gravity theories as particular cases. Subsequently, it may be used as a powerful tool to compare independent models from a global perspective. The authors of [4] use this approach to describe the three modified gravity theories in terms of the same global variables, providing a compact, visual comparison between them. Again, replicating the derivation here and interpreting the conclusions would only make sense if we had properly introduced the modified gravity theories, so we refer to the cited source for the details.

2.2.5 Interacting dark energy

Finally, we address the interesting case of interacting dark energy. Given the intriguing nature of dark energy and dark matter, and the little fundamental knowledge that we have about either species, interaction between both has become another relevant candidate to explain dynamical dark energy, and in particular quintom behaviour. The articles [34, 23] provide comprehensive surveys on the topic, while other relevant works are found in [36, 20]. Interaction with other species like neutrinos has also been proposed [41].

The interacting dark energy model is governed by the following simple system of differential equations (see e.g. [36]),

$$\dot{\rho}_{\text{DE}} + 3H(1+w)\rho_{\text{DE}} = Q_{\text{int}}, \quad (2.35)$$

$$\dot{\rho}_{\text{DM}} + 3H\rho_{\text{DM}} = -Q_{\text{int}}, \quad (2.36)$$

where Q_{int} is the interaction term causing energy transfer between dark energy and dark matter. This term is commonly written in terms of the Hubble parameter and the dark energy and dark matter densities, for instance $Q_{\text{int}} = 3b^2H(\rho_{\text{DE}} + \rho_{\text{DM}})$ in [36].

An alternative model leading to mixture between the dark species can be obtained from a two-field theory in which a scalar field associated to dark energy and a spinor field associated to dark matter are coupled [35].

Using either formalism, the involved parameters can be adjusted in such a way that interacting dark energy models are consistent with the recent observations discussed in Section 1.2. Interestingly, interacting dark energy models are proving to be relevant not only in the study of dynamical dark energy. As an example, a recent estimate of the Hubble constant H_0 using an interacting model seems to reduce the Hubble tension to a level compatible with a statistical fluctuation [11].

2.3 The h-essence model

The h-essence model [37] is a particular case of a two-field model. This theory will play a central role in the work, since the model that we develop in Chapter 3 generalises the Lagrangian arising from h-essence. The authors designed h-essence with the aim of creating a model able to cross the phantom divide without leading to instabilities. As of 2005, this was one of the first works on dynamical dark energy and, specifically, one of the first modified theories to overcome the No-Go Theorem.

We replicate here the derivation of the h-essence model in [37] and its most important properties. The idea of h-essence is to achieve the hyperbolic nature of the quintom model in Eq. (2.20) by considering a single *complex* scalar field,

$$\Phi = \phi_1 + i\phi_2, \quad (2.37)$$

instead of two independent fields. Here, ϕ_1 and ϕ_2 are real fields. In order to produce a hyperbolic field-space metric, we need to express the Lagrangian density in terms of Φ as

$$\mathcal{L} = -\frac{1}{4} [(\partial\Phi)^2 + (\partial\Phi^*)^2] - V(\Phi, \Phi^*), \quad (2.38)$$

instead of the canonical form for a complex field $\Psi = \psi_1 + i\psi_2$, given by

$$\mathcal{L} = -\frac{1}{2}(\partial^\mu\Psi^*)(\partial_\mu\Psi) - V(|\Psi|). \quad (2.39)$$

Like this, Eq. (2.38) reduces to the form of Eq. (2.20) when written in terms of ϕ_1 and ϕ_2 . In contrast, the canonical form would produce a Euclidean field-space metric, with signature (1, 1). The reader should be aware that the original paper [37] uses the opposite sign convention for the spacetime metric, so the kinetic terms have opposite signs in their derivation.

It is worth noting that the canonical Lagrangian in Eq. (2.39) is invariant under any transformation of the class

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \mapsto \begin{pmatrix} \psi_1 \cos \alpha - \psi_2 \sin \alpha \\ \psi_1 \sin \alpha + \psi_2 \cos \alpha \end{pmatrix} \quad (2.40)$$

with α a real number. In other words, the Lagrangian is invariant under rotations of the complex field Ψ , as it ultimately depends on the module $|\Psi|$.

In turn, the kinetic part of the hyperbolic Lagrangian from Eq. (2.38) is invariant under hyperbolic rotations:

$$\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \mapsto \begin{pmatrix} \phi_1 \cosh \alpha - i\phi_2 \sinh \alpha \\ -i\phi_1 \sinh \alpha + \phi_2 \cosh \alpha \end{pmatrix}. \quad (2.41)$$

Observe that this transformation preserves $\phi_1^2 - \phi_2^2$, in the same way that Eq. (2.40) preserved $|\Psi|^2 = \psi_1^2 + \psi_2^2$. Hence, if we demand the potential

in Eq. (2.38) to depend only on $\phi_1^2 - \phi_2^2$, the non-canonical Lagrangian associated to the field Φ will be invariant under the class of transformations given by Eq. (2.41). This invariance under *hyperbolic* transformations explains the name *h-essence*, using the common suffix for dark energy species.

Now, let us introduce a suitable pair of variables that will replace ϕ_1 and ϕ_2 as the main fields of h-essence. First, we make the following observation about the canonical complex field from Eq. (2.39). In a rotationally invariant setting, it is natural to employ polar coordinates (ψ, η) to describe the complex field $\Psi = \psi e^{i\eta}$, since transformations of the form of Eq. (2.40) can be written simply as

$$\psi \mapsto \psi, \quad \eta \mapsto \eta + \alpha. \quad (2.42)$$

We may define analogous coordinates (ϕ, θ) in the hyperbolic case, using the corresponding expressions

$$\phi^2 = \phi_1^2 - \phi_2^2 \quad \text{and} \quad \theta = \tanh^{-1} \left(\frac{\phi_2}{\phi_1} \right). \quad (2.43)$$

Then, the components of Φ are written “polar” form as

$$\phi_1 = \phi \cosh \theta, \quad \phi_2 = \phi \sinh \theta. \quad (2.44)$$

Finally, with this description, the hyperbolic Lagrangian is invariant to rotations of the form

$$\phi \mapsto \phi, \quad \theta \mapsto \theta + \alpha. \quad (2.45)$$

Thus, we observe that the fundamental idea of h-essence is simply to write the two-field quintom Lagrangian from Eq. (2.20) in the hyperbolic version of polar coordinates. Let us see how this Lagrangian looks like in terms of the new coordinates ϕ and θ . Starting from Eq. (2.38), we have

$$\begin{aligned} & -\frac{1}{4} [(\partial\Phi)^2 + (\partial\Phi^*)^2] - V(\Phi, \Phi^*) \\ &= -\frac{1}{2} [(\partial\phi_1)^2 - (\partial\phi_2)^2] - V(\phi_1^2 - \phi_2^2) \\ &= -\frac{1}{2} [(\partial(\phi \cosh \theta))^2 - (\partial(\phi \sinh \theta))^2] - V(\phi) \\ &= -\frac{1}{2} [(\partial\phi)^2 \cosh^2 \theta + \phi^2 (\partial\theta)^2 \sinh^2 \theta \\ & \quad - (\partial\phi)^2 \sinh^2 \theta - \phi^2 (\partial\theta)^2 \cosh^2 \theta] - V(\phi). \end{aligned} \quad (2.46)$$

This simplifies to

$$\mathcal{L} = -\frac{1}{2} [(\partial\phi)^2 - \phi^2 (\partial\theta)^2] - V(\phi), \quad (2.47)$$

which is the expression of the Lagrangian of h-essence that we will use. Observe that, under the general framework of the field-space metric γ^{jk}

from Section 2.2.1, the Lagrangian in Eq. (2.47) belongs to the same class as the basic two-field quintom Lagrangian: the metric has signature $(1, -1)$ at every point of the field space (assuming $\phi \neq 0$). The change of coordinates will certainly help us understand the system and simulate its evolution numerically, but the analysis in terms of the metric γ^{jk} reveals that the underlying field space has exactly the same form.

2.3.1 Dynamics of h-essence

In this occasion, we are not only interested in writing the cosmological observables in terms of the involved fields, but we also wish to derive the equations of motion for ϕ and θ . This is because we will later calculate the cosmic evolution of the main observables using a model that generalises h-essence, and the dynamics is essentially the same for both models.

Recall that we work in flat spacetime with the FLRW metric,

$$ds^2 = -dt^2 + g_{ij}dx^i dx^j, \quad (2.48)$$

where $g_{ij} = a(t)^2 \delta_{ij}$ for $i, j \in \{1, 2, 3\}$.

Let us calculate the Euler-Lagrange equations associated to the Lagrangian of h-essence, which take the form

$$\frac{\partial \mathcal{L}}{\partial \sigma} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \sigma)} \right), \quad (2.49)$$

with $\sigma \in \{\phi, \theta\}$.

For $\sigma = \phi$, the left-hand side reads

$$\frac{\partial \mathcal{L}}{\partial \phi} = \phi(\partial\theta)^2 - V'(\phi), \quad (2.50)$$

where we write $V'(\phi)$ for the total derivative $dV/d\phi$. For the right-hand side, we have

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) &= -\partial_\mu (\partial^\mu \phi) \stackrel{\text{def}}{=} -\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi) \\ &= -\frac{1}{a^3} \partial_\mu (a^3 g^{\mu\nu} \partial_\nu \phi) = -\frac{1}{a^3} (\partial_\mu a^3) g^{\mu\nu} \partial_\nu \phi - g^{\mu\nu} \partial_\mu \partial_\nu \phi, \end{aligned} \quad (2.51)$$

where we have used $\partial_\mu g^{\mu\nu} = 0$ since the only non-constant term is $a(t)$ in the $\mu > 0$ coordinates. Again, given that the scale factor is a function of t , only the $\mu = 0$ component survives the term $(\partial_\mu a^3)$. Hence, using that the metric is diagonal, the expression (2.51) reduces to

$$-3\frac{\dot{a}}{a} g^{00} \partial_0 \phi - g^{\mu\nu} \partial_\mu \partial_\nu \phi = 3H\dot{\phi} + \ddot{\phi} - \frac{\nabla^2 \phi}{a^2}. \quad (2.52)$$

We are left with the equation of motion for the field ϕ ,

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\nabla^2\phi}{a^2} - \phi(\partial\theta)^2 + V'(\phi) = 0. \quad (2.53)$$

In the case of θ , the left-hand side vanishes as there is no explicit dependence $\mathcal{L}(\theta)$. In the right-hand side, we have

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \theta)} \right) = \partial_\mu (\phi^2 \partial^\mu \theta) = \partial_\mu (\phi^2) \partial^\mu \theta + \phi^2 \partial_\mu (\partial^\mu \theta). \quad (2.54)$$

With a similar reasoning as above, this becomes

$$2\phi \partial_\mu \phi \partial^\mu \theta - 3H\phi^2 \dot{\theta} - \phi^2 \left(\ddot{\theta} - \frac{\nabla^2 \theta}{a^2} \right). \quad (2.55)$$

Hence, after rearranging and replacing with the FLRW metric, we obtain the equation of motion for the field θ :

$$\phi^2 \ddot{\theta} + (3H\phi^2 + 2\phi\dot{\phi})\dot{\theta} - \phi^2 \frac{\nabla^2 \theta}{a^2} - \frac{2\phi}{a^2} \partial_i \phi \partial_i \theta = 0. \quad (2.56)$$

Furthermore, if we assume that both ϕ and θ are homogeneous, all spatial derivatives vanish, and the equations of motion simplify to

$$\ddot{\phi} + 3H\dot{\phi} + \phi\dot{\theta}^2 + V'(\phi) = 0, \quad (2.57)$$

$$\phi^2 \ddot{\theta} + (3H\phi^2 + 2\phi\dot{\phi})\dot{\theta} = 0. \quad (2.58)$$

Now, we can make an interesting remark on Eq. (2.58). Multiplying the equation by a^3 , we obtain

$$0 = a^3 \phi^2 \ddot{\theta} + 3a^2 \dot{a} \phi^2 \dot{\theta} + a^3 2\phi\dot{\phi}\dot{\theta} = \frac{d}{dt} \left(a^3 \phi^2 \dot{\theta} \right). \quad (2.59)$$

This means that the quantity

$$Q = a^3 \phi^2 \dot{\theta} \quad (2.60)$$

is a constant of motion. Although it has come almost magically from the derivation, this should be no surprise to an attentive reader. First, recall that ϕ and θ are the hyperbolic analogue of polar coordinates in Euclidean space. Thus, the quantity $\phi^2 \dot{\theta}$ mimics $r^2 \dot{\varphi}$ in mechanics, which is no more than the angular momentum of the system. Hence, Eq. (2.59) can be regarded as the angular momentum conservation law for our hyperbolic two-field model. Second, note that this conservation law emerges from the rotationally invariant nature of the Lagrangian of a rotating system, as a consequence of Noether's theorem (see [21, p. 378-379]). In our case, we have the exact same invariance under translations of the coordinate θ , and thus the corresponding

conservation law follows. The presence of an additional a^3 factor is due to the FLRW spacetime metric and the multiplicities of ϕ and $\dot{\theta}$.

Using the generalised angular momentum Q , we can replace $\dot{\theta}$ in Eq. (2.57) and eliminate the dependence on the field ϕ ,

$$\ddot{\phi} + 3H\dot{\phi} + \frac{Q^2}{a^6\phi^3} + V'(\phi) = 0. \quad (2.61)$$

Next, let us connect this derivation with the observables we are interested in. From the general expressions for multi-field models in Eqs. (2.24) and (2.23), we may write density and pressure in h-essence as

$$\rho = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\frac{Q^2}{a^6\phi^2} + V(\phi), \quad (2.62)$$

$$p = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\frac{Q^2}{a^6\phi^2} - V(\phi). \quad (2.63)$$

Note that the term associated to $\dot{\theta}^2$ is now different, as it replaces $\phi^2\dot{\theta}^2/2$ in the Lagrangian instead of $\phi\dot{\theta}^2$, obtained after taking derivatives. The equation of state is, hence,

$$w = \frac{\dot{\phi}^2 - Q^2/a^6\phi^2 - 2V(\phi)}{\dot{\phi}^2 - Q^2/a^6\phi^2 + 2V(\phi)}. \quad (2.64)$$

Finally, to be able to solve Eq. (2.61), we need to obtain the only ingredient whose expression we do not know yet: the Hubble parameter. We are mainly interested in the behaviour of dark energy at low redshift, as this is the epoch where observations show evidence of deviation from the concordance model (see Section 1.2). Therefore, we will neglect the effects of radiation and other non-abundant species at the present time, and consider a total energy density

$$\rho_{\text{tot}} = \rho_m + \rho_{\text{DE}}. \quad (2.65)$$

With that, the first Friedmann equation (Eq. (1.4)) in the h-essence model is given by

$$H^2 = \frac{1}{3}(\rho_m + \rho_{\text{DE}}) = \frac{1}{3}\left(\rho_{m,0}a^{-3} + \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\frac{Q^2}{a^6\phi^2} + V(\phi)\right), \quad (2.66)$$

where $\rho_{m,0}$ denotes the matter density ρ_m at $a_0 = 1$.

2.3.2 Numerical results

In the previous section, we derived the dynamics of h-essence and reached the equation of motion of the system, Eq. (2.61), which is ready to be integrated. Here, we will discuss the approach taken in the original paper [37] and

comment on the results they obtained, evaluating their significance in the context of this work.

A first thing to note is that the analytical integration of Eq. (2.61) is not realistic, especially as it involves the complex expression of $H = \dot{a}/a$, which depends on ϕ , $\dot{\phi}$ and $V(\phi)$. Therefore, numerical integration is required.

In [37], the authors carry out a canonical transformation of the second-order differential equation (2.61) into a system of first-order equations, by defining

$$\chi = \dot{\phi}. \quad (2.67)$$

Together with this change, it turns out to be really helpful to write the system using the scale factor a as the integration variable instead of the cosmic time t . To this end, we calculate

$$\frac{d\chi}{da} = \frac{1}{\dot{a}} \frac{d\dot{\phi}}{dt} = \frac{1}{aH} \ddot{\phi} = -\frac{1}{aH} \left[3H\chi + \frac{Q^2}{a^6\phi^3} + V'(\phi) \right], \quad (2.68)$$

whereas

$$\frac{d\phi}{da} = \frac{1}{\dot{a}} \frac{d\phi}{dt} = \frac{\chi}{aH}. \quad (2.69)$$

At this point, we are ready to carry out the numerical integration. Since the numerical scheme used in Chapter 3 is a generalisation of h-essence, we have decided to integrate the h-essence equation ourselves with that method, aiming to replicate the results in [37]. In addition, this has allowed us to extract more observables from the model, namely the energy densities of matter and dark energy, which were not published in the original paper.

Fig. 3 shows the plots obtained for the evolution of the field ϕ and its derivative χ , the equation of state, and the matter and dark energy densities. The authors of [37] did not provide the initial conditions set in their integration, nor the precise values used for certain constants such as H_0 . Thus, we needed to figure them out from the published plots, and we cannot expect to faithfully reproduce the exact curves in their article. They did specify the constant $Q = 1$ and the potential $V(\phi) = 5\phi^4$ (that is, $\lambda = 5$ and $\alpha = 4$ in Eq. (2.2)). In our integration, we keep these three values.

In any case, the qualitative behaviour of our results does match the one in [37]. The main goal of crossing the phantom divide is achieved smoothly, as the equation of state is located around $w = 0$ for the initial and final scale factors and reaches a minimum of $w \approx -1.3$ at $a \approx 1$. The minimum coincides roughly with the maximum of ϕ , where $\chi \approx 0$. Looking at Eq. (2.64), and assuming the slow-roll regime, for $\dot{\phi} \approx 0$ and $a \approx 1$ the function $w(\phi)$ behaves approximately as

$$w \approx \frac{\varepsilon/\phi^2 + 1}{\varepsilon/\phi^2 - 1} = \frac{\varepsilon + \phi^2}{\varepsilon - \phi^2}, \quad (2.70)$$

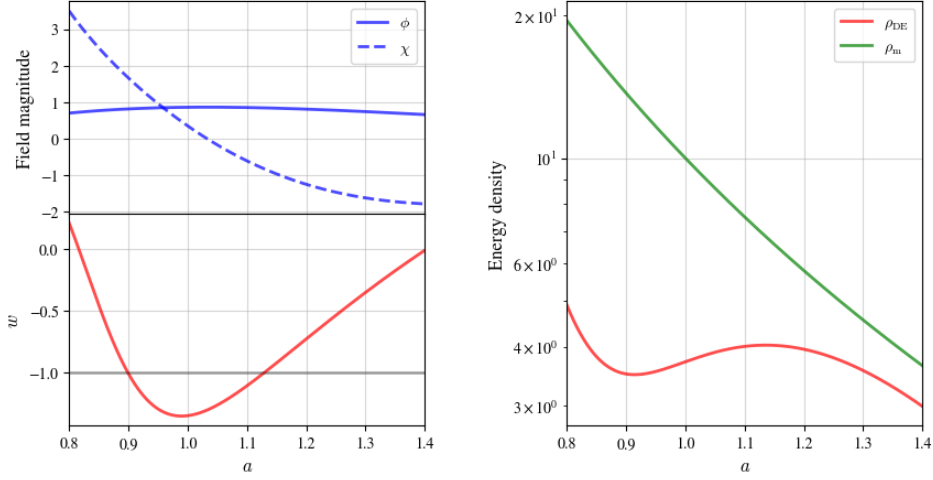


Figure 3: Results of numerical integration of the h-essence model using $Q = 1$, $\rho_{m,0} = 10$, $\lambda = 8$ and $\alpha = 4$. The left panel shows the evolution of the field ϕ , its derivative χ , and the equation of state w with respect to the scale factor a . The phantom divide is depicted as a grey horizontal line. The right panel shows the evolution of the energy densities of matter and dark energy. Initial values were $a_i = 0.8$, $\phi_i = 0.7$ and $\dot{\phi}_i = 3.5$.

with $\varepsilon \ll 1$. Taking derivatives (we use primes for an arbitrary derivative),

$$w' \approx \frac{4\phi\phi'\varepsilon}{(\varepsilon - \phi^2)^2}. \quad (2.71)$$

Then, indeed, we expect extrema of w and ϕ to happen approximately at the same time.

Regarding the crossing points, they come at $a \approx 0.9$ and $a \approx 1.1$. We observe on the left panel of Fig. 3 that χ (and even more χ^2) varies reasonably faster than ϕ , with χ^2 going well beyond 1 already at the cited scale factors. Since $Q^2/a^6\phi^3$ is of the order of 1, we expect the quintessence term χ^2 to dominate the kinetic part in Eq. (2.64) outside the interval $[0.9, 1.1]$, and thus find phantom behaviour only within those times.

It is left to address the behaviour of energy densities. Since the relation $\rho_m(a) \sim a^{-3}$ was set as a condition, the two interesting features of this plot are the ratio of energy densities and the shape of the curve $\rho_{DE}(a)$. The first is quite disappointing, as we expect dark energy to be dominant in the present epoch by roughly a factor 2, but the model yields a smaller density over the whole range, with notably $\rho_{DE,0}/\rho_{m,0} \approx 0.4$. As for the shape, we observe a correlation between the regime of w and the dark energy density, which only increases when the equation of state lies within the phantom regime. Thus, the model does simulate correctly the dependence $\rho_{DE}(a)$ from Eq. (1.9).

In any case, the purpose of [37] was not to produce a realistic theory

of dynamical dark energy, but just to create a model able to elude the No-Go Theorem and prove that a basic two-field model can suffice to cross the phantom divide smoothly. Hence, it did not make much sense for the authors to vary the parameters and initial conditions until reaching a realistic solution. In the following chapter, we will develop a more general model and look for variations of h-essence that do yield a more realistic evolution both for w and ρ_{DE} . Specifically, our numerical approach differs from that of h-essence in two ways. On the one hand, we carry out simulations that go further in the past, in order to compare the results of the simulations with the observational data from Section 1.2 at higher redshifts. On the other hand, we aim for the equation of state to behave locally like the best-fit approximations of the w_0w_a CDM model, and to cross the phantom divide from phantom to quintessence at a reasonable scale factor. In other words, while being aware that building a complete cosmological model relying on a two-field quintom theory goes beyond the scope of this work, we attempt to reproduce the cited properties of dark energy as realistically as possible.

3 A new model: s-essence

In the previous chapter, we reviewed the most important models of quintom dark energy that aim to overcome the No-Go Theorem and allow the equation of state to cross the phantom divide. Concretely, we presented the details of h-essence, a theory that studies the simplest two-field model using a suitable choice of coordinates and provides solutions that cross the cosmological constant boundary without divergences.

Most of those models, including h-essence, were content with avoiding the instabilities at the phantom divide and did not care much about producing realistic solutions for the cosmological observables computed. In fact, the first instances of such theories were published shortly after evidence for dark energy was obtained in 1998. The precision and scope of cosmological observations have evolved vastly since then, and accurate estimates of the relevant parameters in dynamical dark energy were obtained only recently (see Section 1.2), leaving little room for the precise fitting of these models.

In this chapter, we extend the theory of h-essence to a more general model, and then attempt to reproduce observational data by varying the parameters involved. Specifically, we seek to produce a realistic evolution of the equation of state w and the dark energy density ρ_{DE} , taking as a benchmark the estimates in Figs. 1 and 2. Whenever we do not manage to achieve results consistent with the data, we try to understand whether the equations prevent the model to yield those values.

The main idea of our model is to replace the ϕ^2 factor in the Lagrangian of h-essence (Eq. (2.47)) by an arbitrary function $f(\phi)$. In models like h-essence, the phantom divide is crossed upward (resp. downward) when the kinetic term $\dot{\theta}^2$ of the phantom field becomes smaller (larger) than that of quintessence, $\dot{\phi}^2$. Our first idea was that, using an arbitrary function f with no definite sign, this transition can also be achieved when f changes its sign, possibly leading to a novel behaviour in the nature of the crossing. Due to the relevant role of the sign of f in the model, we decided to name it *s-essence*, mimicking the terminology used with previous dark energy candidates.

Unfortunately, when studying the dynamics of s-essence one finds out that f cannot change its sign smoothly. Indeed, in the next section we show that s-essence also has a generalised angular momentum which is conserved, given by $Q = a^3 f(\phi) \dot{\theta}$ (see Eq. (3.8)). Thus, except in the trivial case $Q = 0$ where the phantom term disappears, the function $f(\phi)$ can never cross 0 in a continuous way. In other words, we have a “local” version of the No-Go Theorem for s-essence and the $f = 0$ divide. We discuss minimal changes to s-essence which might allow this crossing in the conclusions of Chapter 4.

With all that, the results that we have obtained using s-essence are significantly better than the results of h-essence reviewed before. In order to present them, we follow an analogous approach as for h-essence: starting from the modified Lagrangian, we will derive the equations of motion and

then integrate them numerically. Much of the work is already done, since the structure of the equations is essentially identical once we take care of the function f , and this depends only on ϕ just like the factor ϕ^2 did. Hence, we will omit most steps in the derivations, and focus on the role of f and its effects on the results of the integration.

3.1 Dynamics of s-essence

Thus, let us start by considering a Lagrangian like the one in Eq. (2.47) and replace the ϕ^2 factor by an arbitrary function $f(\phi)$:

$$\mathcal{L} = \frac{1}{2} [(\partial\phi)^2 - f(\phi)(\partial\theta)^2] - V(\phi) \quad (3.1)$$

Since the sign of $f(\phi)$ is, a priori, not determined, this Lagrangian is not constrained within any class of models with respect to the metric γ^{jk} from Eq. (2.21). In other words, it might oscillate between the signatures (1, 1) and (1, -1), switching its behaviour from two-field quintessence to quintom.

To obtain the equations of motion, it suffices to repeat the derivation done for h-essence in Chapter 2 starting from the Euler-Lagrange equations, now using $f(\phi)$ in the place of ϕ^2 . One reaches the following evolution equations for ϕ and θ :

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\nabla^2\phi}{a^2} - \frac{1}{2}f'(\phi)(\partial_\mu\theta)^2 + V'(\phi) = 0, \quad (3.2)$$

$$f(\phi)\ddot{\theta} + \left(3Hf(\phi) + f'(\phi)\dot{\phi}\right)\dot{\theta} - f(\phi)\frac{\nabla^2\theta}{a^2} - f'(\phi)\partial_i\phi\partial_i\theta = 0. \quad (3.3)$$

Again, assuming homogeneity of ϕ and θ , these become

$$\ddot{\phi} + 3H\dot{\phi} - \frac{1}{2}f'(\phi)\dot{\theta}^2 + V'(\phi) = 0, \quad (3.4)$$

$$f(\phi)\ddot{\theta} + \left(3Hf(\phi) + f'(\phi)\dot{\phi}\right)\dot{\theta} = 0. \quad (3.5)$$

Unlike h-essence, this model does not emerge from a coordinate transformation of Eq. (2.20), and does not even belong to the hyperbolic class of signatures of γ^{jk} , as discussed above. Nonetheless, the Lagrangian in Eq. (3.1) is still invariant with respect to hyperbolic rotations $\theta \mapsto \theta + \alpha$. Indeed, this is clear as the only term involving θ is the derivative $(\partial\theta)^2$. According to Noether's theorem, this symmetry leads to a conserved quantity Q , a sort of generalised angular momentum as in the case of h-essence. In fact, multiplying Eq. (3.5) by a^3 , we obtain

$$a^3f(\phi)\ddot{\theta} + 3\dot{a}a^2f(\phi)\dot{\theta} + f'(\phi)\dot{\phi}\dot{\theta}a^3 = 0, \quad (3.6)$$

or equivalently

$$\frac{d}{dt} \left(a^3 f(\phi) \dot{\theta} \right) = 0. \quad (3.7)$$

Hence, the generalised angular momentum, which is a constant of motion of s-essence, is now given by

$$Q = a^3 f(\phi) \dot{\theta}. \quad (3.8)$$

Observe that this expression coincides with Eq. (2.60) up to the replacement of ϕ^2 by the generic function $f(\phi)$. Furthermore, note that if $Q = 0$, then $\dot{\theta} = 0$ and the s-essence Lagrangian in Eq. (3.1) reduces to the quintessence Lagrangian. Otherwise, none of the factors on the right-hand side can become 0, and in particular the function $f(\phi)$ has a definite sign. Hence, in practice, each specific instance of s-essence does belong to a certain class with respect to the signature of the field metric γ^{jk} , depending on the initial value of $f(\phi)$.

Using the constant quantity Q , we can eliminate $\dot{\theta}$ from Eq. (3.4), which becomes

$$\ddot{\phi} + 3H\dot{\phi} + \frac{1}{2} \frac{Q^2 f'(\phi)}{a^6 f(\phi)^2} + V'(\phi) = 0. \quad (3.9)$$

The main cosmological observables can also be put in terms of the generalised angular momentum Q . Recall that, assuming dark energy is a perfect fluid and the fields are homogeneous, the energy density and pressure can be written as

$$\rho = \frac{1}{2} \left[\dot{\phi}^2 - \frac{Q^2}{a^6 f(\phi)} \right] + V(\phi), \quad (3.10)$$

$$p = \frac{1}{2} \left[\dot{\phi}^2 - \frac{Q^2}{a^6 f(\phi)} \right] - V(\phi). \quad (3.11)$$

Then, the equation of state is given by

$$w = \frac{\dot{\phi}^2 - Q^2/a^6 f(\phi) - 2V(\phi)}{\dot{\phi}^2 - Q^2/a^6 f(\phi) + 2V(\phi)}. \quad (3.12)$$

Finally, the Hubble parameter evolves as (compare with Eq. (2.66))

$$H^2 = \frac{1}{3} \left(\rho_{m,0} a^{-3} + \frac{1}{2} \dot{\phi}^2 - \frac{Q^2}{2a^6 f(\phi)} + V(\phi) \right). \quad (3.13)$$

Before continuing with the numerical integration of Eq. (3.9), we need to give a specific form to the free functions $f(\phi)$ and $V(\phi)$. First, recall that in h-essence we used $f(\phi) = \phi^2$ and $V(\phi) = 5\phi^4$. For the potential, we will mostly work with the general form $V(\phi) = \lambda\phi^\alpha$ from Eq. (2.2) and vary the values of λ and more notably α across different simulations. We will show that the value of α can have relevant implications on the behaviour of the system, and a negative α can effectively prevent the model from crossing the phantom divide in certain settings. Additionally, we will set $V(\phi) = \lambda \cosh(\phi)$ in one of the simulations, which constrains ϕ to smaller values and gives rise to a different qualitative behaviour.

Regarding the function f , recall that we initially intended it to be sign-indefinite. After the observation that f is not allowed to change signs in s-essence, this requirement does not make sense any more. Nonetheless, one of our first candidates subject to this requirement was the function $f(\phi) = \tanh(\phi/\phi_0)$, which turned out to give very good results for several choices of parameters, and we decided to keep it in all of our simulations. It also has the desirable properties of being bounded and bijective, in such a way that one can easily estimate the value of $f(\phi)$ given its argument and vice versa. Finally, $f(\phi)$ has always the same sign as ϕ . Thus, since $f(\phi)$ is not allowed to change signs, the initial sign of ϕ will determine whether the model describes quintom or quintessence dark energy. Here, ϕ_0 is a real, positive constant. After carrying out a series of simulations, we found out that its value did not notably change the evolution of the system, and decided to set $\phi_0 = 1$ in all the results presented here.

3.2 Numerical integration

We move on to the numerical solving of Eq. (3.9), from which we will discuss the results of our model. The analytical integration of the equation of motion is now even less feasible than in h-essence, since there is an additional dependence on the function $f(\phi)$ and its derivative $f'(\phi)$, and $f(\phi)$ is also present in the expression of H^2 . Even replacing $f(\phi) = \tanh(\phi)$ does not simplify things much. Thus, we will follow a numerical scheme based on the one used for h-essence. Again, we skip most details as they are mainly coincident.

Let us again set $\chi = \dot{\phi}$ and replace the cosmic time by the scale factor as the integration variable. The equivalent expressions to Eqs. (2.68) and (2.69) are

$$\frac{d\chi}{da} = -\frac{1}{aH} \left[3H\chi + \frac{1}{2} \frac{Q^2 f'(\phi)}{a^6 f(\phi)^2} + V'(\phi) \right], \quad (3.14)$$

$$\frac{d\phi}{da} = \frac{\chi}{aH}. \quad (3.15)$$

For the integration, we used the Runge-Kutta-Fehlberg method (RKF45) implemented within the Python function `solve_ivp` from `scipy.integrate`. In total, we solved the equation for five different sets of values for the parameters of the model, listed in Table 2. In this way, each set j gives rise to a different model within the class of s-essence, that we will refer to as Model j . Sets 0 to 3 are ordered roughly by their success at reproducing the observational data, whereas Set 4 is analysed later as it features a different potential. In particular, $V(\phi) = \lambda\phi^\alpha$ in the first four and $V(\phi) = \lambda \cosh(\phi)$ in Set 4.

Note that the integration is carried out forward ($a_i < a_f$) in Sets 1 to 4 but backward in Set 0. This is because controlling the initial conditions at

Set	Q	$\rho_{m,0}$	λ	α	(a_i, a_f)	ϕ_i	$\dot{\phi}_i$
0	0.01	4.0	5.0	-0.1	(1.0, 0.15)	0.1	1.0
1	0.01	4.0	5.0	4.0	(0.2, 1.0)	0.2	3.0
2	0.05	4.0	5.0	-2.0	(0.3, 1.0)	-1.4	3.5
3	0.08	4.0	5.0	-2.0	(0.3, 1.0)	1.4	-3.5
4	0.01	4.0	1.0	-	(0.15, 1.0)	3.0	-1.0

Table 2: Sets of values for the parameters of s-essence used in the simulations. Sets 0 to 3 have a potential $V(\phi) = \lambda\phi^\alpha$, whereas Set 4 has $V(\phi) = \lambda \cosh(\phi)$. Subscripts i and f indicate the initial and final values of the variables in the integration.

$a_0 = 1$ was helpful in some cases, while ensuring stable conditions in the far past ($a \ll 1$) was more important in others.

3.3 Results

In the next pages, we present the results obtained for each set and comment on the influence of the involved parameters. Specifically, we motivate why each choice of parameters and initial conditions is relevant for the work and relate the observed results to the expected behaviour of the analytical expressions of Section 3.1. A common feature of all sets is the low value of Q . On the one hand, the following inequality must hold at all times for the dark energy density to remain positive (see Eq. (3.10)):

$$Q^2 \leq a^6 f(\phi) \left(\dot{\phi}^2 + 2V(\phi) \right). \quad (3.16)$$

This imposes an upper bound on Q depending on the exact shape of $V(\phi)$. On the other hand, notice that if Q is sufficiently small, we can approximate the equation of state at $a \approx 1$ by (see Eq. (3.12))

$$w \approx \frac{\dot{\phi}^2 - \varepsilon/f(\phi) - 2V(\phi)}{\dot{\phi}^2 - \varepsilon/f(\phi) + 2V(\phi)}, \quad (3.17)$$

where $\varepsilon \ll 1$. Hence, as long as $f(\phi)$ does not approach 0, near the present time the model behaves essentially as a one-field quintessence model, matching recent observations which estimate $w_0 > -1$. In contrast, as the scale factor decreases towards the past, the term $1/a^6$ grows drastically, allowing the crossing of the phantom divide as the negative kinetic term dominates.

For every set, we plot the evolution of the field ϕ and its derivative χ , the energy densities ρ_m and ρ_{DE} , and the equation of state w . As a reference, we include the uncertainty regions of ρ_{DE} (relative) and w at 1σ and 2σ , taken from the right column of Fig. 2c, which correspond to the data combination of

DESI, CMB and Pantheon+. Since the exact plotted data are not published, the regions we draw are an approximation. Furthermore, we also include the 95% uncertainty region around the best-fit line of the w_0w_a CDM model, estimated from Fig. 1 (right panel), again for the combination of DESI DR2, CMB and Pantheon+.

Set 0: consistent with data for $a > 0.8$

Model 0 features a nearly constant potential with $\alpha = -0.1$. Thus, for non-extreme values of ϕ , the potential sits around $V \equiv 5$. Since $Q = 0.01$, for $a \lesssim 1$ we can approximate the equation of state by

$$w \approx \frac{\dot{\phi}^2 - 10^{-4}/f(\phi) - 10}{\dot{\phi}^2 - 10^{-4}/f(\phi) + 10} \approx \frac{\dot{\phi}^2 - 10}{\dot{\phi}^2 + 10}. \quad (3.18)$$

This is the typical behaviour of a quintessence field, and we observe in Fig. 4 that the simulation does agree with this observation. Given that $d\chi/da$ is positive, the equation of state decreases until $a \approx 0.8$.

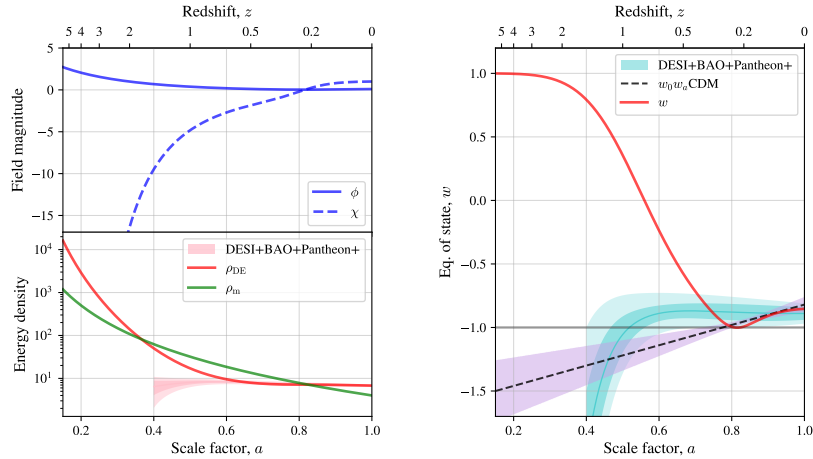


Figure 4: Evolution of the fields ϕ and χ (top left), the dark energy and matter energy densities (bottom left) and the equation of state (right) for the parameters in Set 0 (see Table 2). For comparison, we plot the 1σ and 2σ uncertainty regions of ρ_{DE} and w from Fig. 2c (combined data from DESI+BAO+Pantheon+) and the 95% uncertainty region around the best-fit line of the w_0w_a CDM model from Fig. 1, again using DESI+BAO+Pantheon+.

Then, below that scale factor, the phantom term becomes relevant. However, at the same time, the positive kinetic term was pushing the field towards the potential barrier at $\phi = 0$. When χ vanishes, the field bounces back and the potential accelerates it to larger values, making $\dot{\phi}^2$ important

again. The combination of large quintessence and phantom terms dominate the potential, making w bounce back toward $w = 1$. Although it is normally reached at larger redshifts, this unrealistic equation of state is a common feature of quintom models in the $a \rightarrow 0$ limit (see [18], [34] or h-essence itself [37]), provoked by the dependence of the phantom term on a negative power of a .

With all that, the slope of the equation of state near $a = 1$ is consistent with the linear approximation and lies within the uncertainty region of the reconstructed function using Gaussian process regression, and remains so until touching the phantom divide. More importantly, the simulation yields $\rho_{\text{DE},0}/\rho_{m,0} \approx 1.7$, which is reasonably close to the well-established measured value of $0.69/0.31 \approx 2.2$, especially when comparing with other sets. The curve $\rho_{\text{DE}}(a)$ remains within the uncertainty region of the reconstructed function, and crosses the curve ρ_m at $a_{\text{eq}} \approx 0.85$, which is again consistent with observations. Hence, despite the caveat of not crossing the phantom divide, this model excels at reproducing the behaviour of both w and ρ_{DE} at small redshifts, where the observational data are more accurate.

Set 1: consistent with reconstructions of $w(a)$ up to $z \approx 1$.

Model 1, instead, uses the same potential as h-essence, namely $V(\phi) = \lambda\phi^4$, with $\lambda = 5$ in this case. Now, the potential constrains ϕ to stay within the polynomial walls, and thus bounds the value of $\dot{\phi}^2$. If we start with a slow-rolling field ϕ , we expect quintessence behaviour at small redshift, exactly as before. The simulation results are shown in Fig. 5 support this reasoning.

The change comes when a becomes smaller. As shows the top left panel of the figure, $\dot{\phi}^2$ becomes negligible when the model crosses the phantom boundary, and thus we can approximate the equation of state by

$$\frac{-\varepsilon - 1}{-\varepsilon + 1} \quad (3.19)$$

for $\varepsilon \ll 1$ in a reasonable interval below $a \approx 0.8$, as the model lies in the phantom regime. Beyond that area, the field bounces back and the kinetic term starts to be relevant again. Since the phantom term, driven by a^{-6} , keeps growing, the potential stops dominating the kinetic part and the system approaches $w = -1$ in the high-redshift limit, as discussed for the set 0.

The model yields an approximately linear evolution of the equation of state up to $z \approx 1$, matching very precisely the best-fit line of the $w_0 w_a$ CDM model, and crossing the phantom divide at $a \approx 0.75$, or $z \approx 0.3$. This is consistent with the crossing redshift obtained by reconstruction of w using data from DESI, CMB and Union3 in [10] (see central panel of Fig. 2c). However, it fails to produce a dark energy density compatible with observations, with $\rho_{\text{DE},0}$ lying around 2 orders of magnitude below $\rho_{m,0}$. The

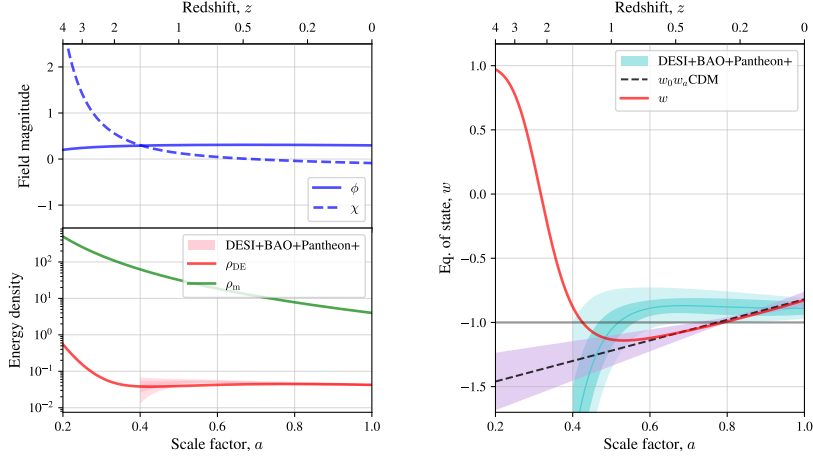


Figure 5: Evolution of the fields ϕ and χ (top left), the dark energy and matter energy densities (bottom left) and the equation of state (right) for the parameters in Set 1 (see Table 2). For comparison, we plot the uncertainty regions obtained from previously published data as in Fig. 4.

curve of evolution of ρ_{DE} does fit within the uncertainty regions considered. In comparison with Model 0, the model given by Set 1 is successful at producing a function $w(a)$ compatible with data up to $z \approx 1$ and crossing the phantom divide, but cannot reproduce the correct values of dark energy density.

Sets 2 and 3: only one can cross the divide

Sets 2 and 3 were chosen to show the local No-Go theorem that prevents $f(\phi)$ from crossing through 0. Note that both share almost all the parameters, the only differences being a small change in Q and the signs of ϕ_i and $\dot{\phi}_i$. The simulation results for both sets are presented in Figs. 6 and 7, respectively.

We observe in the figures that the behaviour of both models is quite similar: the evolution curves of the fields are essentially a mirror reflection of each other, and the dark energy density evolves in a similar way at low redshifts. However, Model 3 crosses the phantom divide in a similar manner as set 1 did, whereas Model 2 stays at the verge but bounces toward $w = -1$ before crossing. In addition to the mathematical argument seen before preventing $f(\phi)$ from switching its sign, the potential used here also pushes ϕ away from 0. Indeed, the function $V(\phi) = \lambda\phi^{-2}$, presents an aggressive wall at $\phi = 0$. For both models, the initial condition (at $a = 0.3$) makes the field advance toward the wall until $\chi = 0$, and then bounce back slowly. In Set 3, since $f(\phi) > 0$, the evolution is analogous to the one described for set 1. Instead, in Set 2 we have $f(\phi) < 0$ and the model

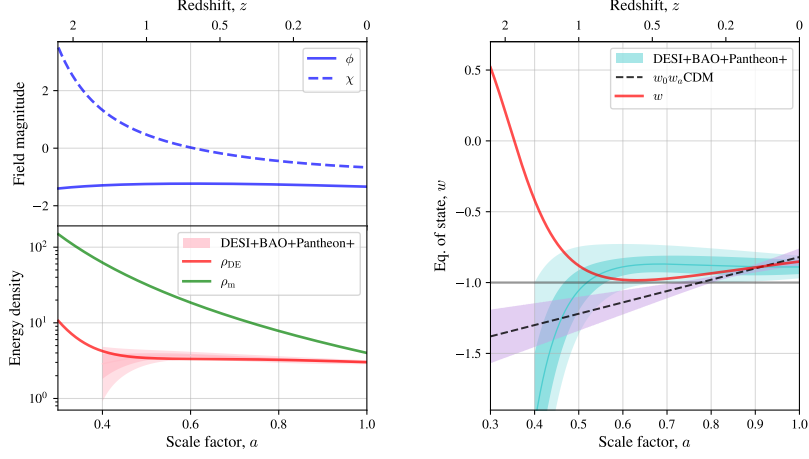


Figure 6: Evolution of the fields ϕ and χ (top left), the dark energy and matter energy densities (bottom left) and the equation of state (right) for the parameters in Set 2 (see Table 2). For comparison, we plot the uncertainty regions obtained from previously published data as in Fig. 4.

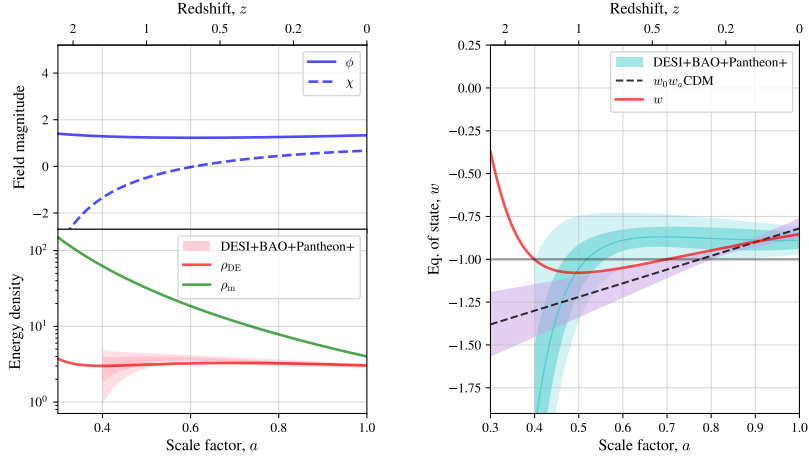


Figure 7: Evolution of the fields ϕ and χ (top left), the dark energy and matter energy densities (bottom left) and the equation of state (right) for the parameters in Set 3 (see Table 2). For comparison, we plot the uncertainty regions obtained from previously published data as in Fig. 4.

behaves just like quintessence, remaining close to the divide when ϕ rolls slowly and approaching $w = -1$ in the high-redshift limit.

For both sets, the dark energy density evolves similarly as for set 1. Now, the value of $\rho_{\text{DE},0}$ is closer to the observations, missing roughly a factor 3 but still lying below $\rho_{m,0}$. The calculated evolution of $\rho_{\text{DE}}(a)$ lies mostly within the relative uncertainty regions of the Gaussian reconstruction process, as it yields an almost constant value with a small bump around $a \approx 0.7$.

Set 4: a steeper potential which yields the best $\rho_{\text{DE}}(a)$

Finally, we consider the set 4, which differs from the rest in that the potential is not as in Eq. (2.2), but instead presents the form $V(\phi) = \lambda \cosh(\phi)$. The simulation results are shown in Fig. 8.

In this model, we observe that the equation of state diverges into the deep phantom regime at moderately high redshifts ($z \approx 5$), while all other models had bounced back toward $w = -1$ already by $z = 2$. The reason for this late rebound is similar to the one discussed for set 1. The hyperbolic cosine potential has exponential walls which are even steeper than those of ϕ^4 , constraining $\dot{\phi}^2$ more strongly. Moreover, the minimum value is now $V(0) = 1$ instead of 0, allowing the potential to dominate both kinetic terms for longer and regardless of the value of ϕ . In any case, simulations reaching larger z did show a violent rebound toward $w = -1$. However, if observational data ever suggest a scenario where $w(a=0) < -1$, a variation of this model may be a very good way to describe it.

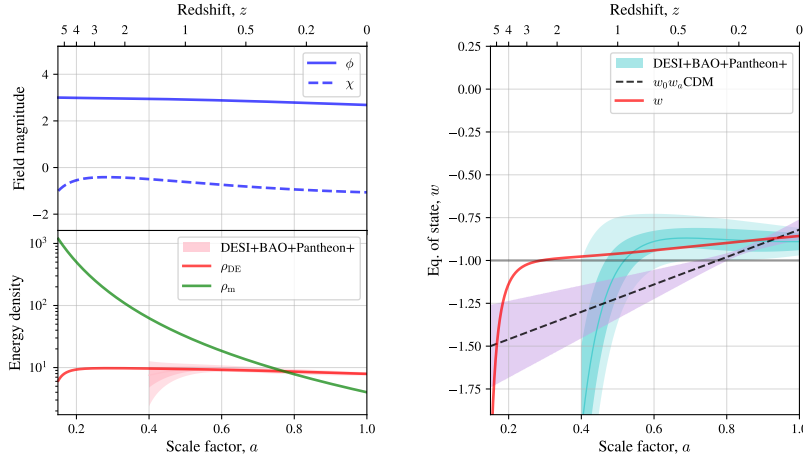


Figure 8: Evolution of the fields ϕ and χ (top left), the dark energy and matter energy densities (bottom left) and the equation of state (right) for the parameters in Set 4 (see Table 2). For comparison, we plot the uncertainty regions obtained from previously published data as in Fig. 4.

Furthermore, set 4 is the one for which the phantom divide is crossed earlier, between $z = 2$ and $z = 3$. This value is compatible with the reconstructed $z_c = 2.18$ from [38] (see Fig. 2a), but lies far from the predictions of the remaining studies from Fig. 2. The slope of the function is also not within the 95% confidence interval of the $w_0 w_a$ CDM model. In turn, the model produces the best value for the energy density ratio, $\rho_{\text{DE},0}/\rho_{m,0} \simeq 2.0$, and the simulation of $\rho_{\text{DE}}(a)$ lies within the 1σ uncertainty interval at all available redshifts.

3.4 Discussion

All in all, we have found instances of s-essence that achieve our primary goals quite successfully. In particular, the models emerging from sets 0, 1 and 4 present each a desirable property that can be exploited further if the parameters are refined or new observations demanding that behaviour are obtained. Table 3 shows a summary of the ability of the models when it comes to producing quintom behaviour, crossing the phantom divide and providing a reasonable dark energy density.

Set	Quintom	Cross $w = -1$	ρ_{DE}
0	No	No	Compatible value and shape
1	A & B	$z \approx 0.3$	~ 100 times too small
2	No	No	~ 3 times too small
3	A & B	$z \approx 0.45$	~ 3 times too small
4	B	$z \approx 2.5$	Compatible value and shape

Table 3: Summary of the properties of the different s-essence models studied.

Model 0 fits the data extremely good up to $z \approx 0.2$, producing an equation of state which basically lies on top of the best-fit line of the $w_0 w_a$ CDM model. Together with that, the value of $\rho_{\text{DE},0}$ is reasonable. With both properties, it is clearly the most successful model at low redshift. However, its almost flat potential prevents the model from crossing the phantom divide despite approaching it closely, failing to yield quintom behaviour.

In that regard, Model 1 partially solves the issue, as it also replicates the local slope of the function $w(a)$ very precisely, and has no problem in crossing the phantom divide at around $z \approx 0.3$ and diving smoothly into the phantom regime. Although it slightly deviates from the Gaussian reconstruction obtained using Pantheon+, it is compatible with the one that uses Union 3, for instance. Hence, this model does a great job at producing a quintom equation of state over a reasonable redshift range. In contrast, its estimate for $\rho_{\text{DE},0}$ lies nowhere near the concordance value.

Lastly, Model 4 gives really good results while using a less standard potential of the form $V(\phi) = \cosh(\phi)$. On the one hand, the initial value and

backward evolution of $\rho_{\text{DE}}(a)$ are the most compatible with the observational data available. On the other, the equation of state evolves reasonably well within the uncertainty margins given by data, and does cross the phantom divide to yield a very negative w at high redshifts. The local slope and the crossing point are not compatible with some of the data, but given the lack of precision beyond small redshifts as of today, it can still be considered a good candidate to describe quintom dark energy.

As we anticipated at the beginning of the results section, all the models are unable to provoke a quintom transition due to a sign flip of the function $f(\phi)$. Indeed, since the (non-zero) constant quantity Q contains $f(\phi)$ as a factor, the function f cannot mathematically become 0 as long as a and $\dot{\phi}$ remain finite. Therefore, if the initial conditions set $\phi_i < 0$, then $f(\phi)$ will remain negative and the system will behave precisely like a two-field quintessence model, without managing to cross the phantom divide at all. Set 2 is an example of this, and in Fig. 6 we observe how the equation of state approaches the $w = -1$ boundary very closely but does not pass through. In turn, all the other models start with $\phi_i > 0$ and reach the phantom regime despite behaving as quintessence for low redshift.

4 Conclusions

In this thesis, we have studied models of dynamical dark energy compatible with the recent observations obtained by DESI [8, 9] and other previous studies, such as [30, 25, 29, 3, 10, 38, 40]. Using combined data of several of these works, independent analyses have been published showing statistical evidence of dynamical dark energy, that is, dark energy with a non-constant density value. These mutually consistent findings pose a challenge for the concordance cosmological model, which has been truly robust so far and managed to explain most observations and the evolution of the universe with great precision.

Modified models have emerged with the aim of producing dynamical dark energy that fits the observations. The most basic such models are the phenomenological w_0w_a CDM model, in which the equation of state $w = -1$ of the cosmological constant is replaced by a linear expression $w(a) = w_0 + w_a(1 - a)$, and the quintessence and phantom one-field models, where (dynamical) dark energy emerges from a real scalar field that rolls with time. However, observations suggest that the dark energy equation of state $w(a)$ crossed the cosmological constant boundary $w = -1$ at some point in the past, exhibiting quintom behaviour, that is, both phantom and quintessence behaviour at different epochs. In particular, a transition from phantom to quintessence is favoured by observations. This creates a big issue for one-field models, since they are mathematically unable to cross the cited boundary, also known as the phantom divide, without fundamental instabilities. Therefore, in order to cross the phantom divide, more complex models are needed. In Section 2.2, we reviewed possible modifications to the quintessence field model that lead to quintom dark energy.

An important instance was the case of h-essence [37]. This approach exploits the invariance of the basic quintom Lagrangian with respect to hyperbolic rotations in order to rewrite the equations of motion in terms of a conserved quantity, which plays the role of a generalised angular momentum. In this way, one may study the evolution of this two-field model in a simplified setting where only the motion of one of the fields needs to be solved, analogous to the study of planetary orbits in terms of their radial distance thanks to the conservation of the angular momentum. The goal of the authors when creating h-essence was to cross the phantom divide smoothly, as this was one of the first models attempting to overcome the No-Go Theorem. Here, in Section 2.3, we reproduced their results and confirmed that h-essence fulfils its main objective.

Nonetheless, with the observational data available today, such models can be tested further and their predictions of multiple observables can be compared with the observations. Hence, our main goal in this thesis was to develop a generalisation of h-essence, and attempt to reproduce at least partially the evolution curves of the dark energy equation of state w and

density ρ_{DE} .

With that aim, we introduced s-essence, by replacing the ϕ^2 factor in the phantom term of the h-essence Lagrangian with an arbitrary function $f(\phi)$. In Section 3.1 we carried out the derivation of the equations of motion for this theory, replicating the steps done for h-essence in Section 2.3, since the structure of both models is essentially identical. As a consequence, we found that s-essence also has a conserved generalised angular momentum, given by $Q = a^3 f(\phi) \dot{\theta}$, which allows to simplify the system of ordinary differential equations describing the motion of the fields.

By construction, the central ingredient of s-essence is the free function $f(\phi)$ which modulates the relevance of the phantom kinetic term in the Lagrangian. Specific forms of this function allow to regard quintessence ($f \equiv 0$), phantom ($f \rightarrow \infty$) or h-essence ($f(\phi) = \phi^2$) as special cases of s-essence. Our primary motivation to introduce such a function in the Lagrangian was to provoke a transition from phantom to quintessence due to a change of sign of f . We pictured this transition as follows: in the past, dark energy would be in the form of phantom as the kinetic phantom term dominates the quintessence term and $f(\phi) > 0$. As f gets smaller, the phantom term becomes less dominant and eventually f switches its sign, in such a way that both kinetic terms are now positive and the system has evolved from the phantom regime into the quintessence divide, crossing the phantom divide as a result of the change of sign of f . The choice $f = \tanh(\phi)$ was made so that $|f| \rightarrow 1$ for large values of ϕ , yielding back usual quintom or two-field quintessence behaviour, while exhibiting a smooth sign change around $\phi = 0$.

Nonetheless, as we discussed in Section 3.1, the conserved quantity Q prevents f from becoming 0, except in the trivial case where $Q = 0$ and s-essence collapses down to quintessence. Therefore, any instance of the presented model is unable to perform the sign transition, unless we allow $\dot{\theta}$ to diverge, or be undefined, in a certain instant. Across the world of physics, there are examples of models with inherent singularities that are nevertheless successful at describing real phenomena. For instance, the centre of a black hole or the plates of a capacitor are singular regions where the gravitational or the electric field are not defined smoothly. A similar solution could be implemented here, by leaving $\dot{\theta}$ undefined at a given time and approach that instant from both sides with solutions exhibiting different signs of $f(\theta)$. However, given that the discontinuity is of essential type, this would in no case yield a smooth evolution for the field θ , and the expressions of observables like w (Eq. (3.12)) and ρ_{DE} (Eq. (3.10)) would still present a divergence. All in all, we end up in a similar situation to that of the No-Go Theorem for the phantom divide.

Thus, alternative solutions need to go beyond the model of s-essence as we have presented it. Observe that the main constraint preventing f to change signs is the conservation law arising from Noether's theorem.

Ultimately, this result can be applied because the Lagrangian in Eq. (3.1) is invariant under translations of the field θ , which behaves as the polar angle of a rotating system. Therefore, removing this symmetry will result in a system without such a conservation law, where f is, a priori, allowed to change signs smoothly. This can be achieved by letting f depend on both ϕ and θ . Hence, in future work one could try symmetric instances in ϕ and θ of the form $f(\phi, \theta) = \tanh(\phi\theta)$ or $f(\phi, \theta) = \tanh(\phi)\tanh(\theta)$, that will lead to more complex expressions of the equations of motion in Eqs. (3.4) and (3.5). Alternatively, one could modify the Lagrangian with a term of the form $\tanh(\theta)e^{-\phi^2}$ which becomes relevant only at small ϕ . In this way, the transition is governed by this new term but the rest of the time the model is equivalent to s-essence, featuring a quasi-conserved quantity Q .

Lastly, let us comment on the results obtained with s-essence. First, we note that the model replicates the success of h-essence at producing dark energy with an equation of state that crosses the phantom divide smoothly. More importantly, our model manages to produce evolution curves for w and ρ_{DE} which are partially consistent with the data. Concretely, we present three successful sets of parameters, each of them exhibiting at least one good property when compared with the data. In this way, Set 0 reproduces very reasonably the expected evolution of both observables at low redshift ($z \lesssim 0.3$). Set 1, in turn, matches closely the linear approximation of the w_0w_a CDM model up to $z \approx 1$. Finally, Set 4, which uses a potential of the form $V = \cosh(\phi)$ instead of the polynomial functions used in the other sets, is able to reproduce the evolution of the energy density very precisely, while crossing the phantom divide at a higher redshift $z \approx 2.5$. The fact that each set excels at a different task is a positive feature for the model. As new observational data will further constrain the evolution of the observables, the results of some sets may no longer lie within the uncertainty margins of the data, but it is likely that some of the three sets will still provide consistent results and be optimised with even more precision.

As we have mentioned before, future work should try slight modifications of the model that do not feature a conserved quantity, or at least not for all times, in such a way that the transition from phantom to quintessence can also be achieved through a sign switch of the function f or a similar factor. Without changing the model, it would also be interesting to automate the search for optimal solutions and perform a quantitative fit of the parameters using (part of) the available data. The main caveat of this approach, and the reason why we did not implement it here, is the a^{-6} factor in the equation of motion. Indeed, for an unlucky choice of parameters, this term can make $\dot{\phi}$ diverge very rapidly as the scale factor becomes smaller, leading to instabilities in the integration. Therefore, in a deeper study of the model, either a robust numerical scheme that carefully addresses overflow issues should be used, or an appropriate region of the parameter space needs to be selected to search for the optimal values.

References

- [1] N. Arkani-Hamed, S. Dubovsky, A. Nicolis, E. Trincherini, G. Villadoro. A measure of de Sitter entropy and eternal inflation. *JHEP*, 05:055, 2007. [arXiv:0704.1814](#).
- [2] C. L. Bennett et al. Nine-Year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Final Maps and Results. *Astrophys. J. Suppl.*, 208, 2012. [arXiv:1212.5225](#).
- [3] D. Brout et al. The Pantheon+ Analysis: Cosmological Constraints. *Astrophys. J.*, 938(2):110, 2022. [arXiv:2202.04077](#).
- [4] Y. Cai, X. Ren, T. Qiu, M. Li, X. Zhang. The Quintom theory of dark energy after DESI DR2. *Nat. Sci. Rev.*, 13(10):nwag115, 2026. [arXiv:2505.24732](#).
- [5] M. Chevallier, D. Polarski. Accelerating universes with scaling dark matter. *Int. J. Mod. Phys. D*, 10:213-224, 2001. [arXiv:gr-qc/0009008](#).
- [6] T. Chiba, T. Okabe, M. Yamaguchi. Kinetically driven quintessence. *Phys. Rev. D*, 62(2), 2000. [arXiv:astro-ph/9912463](#).
- [7] DES Collaboration. The Dark Energy Survey Supernova Program: Light curves and 5-Year data release. *Astrophys. J.*, 975(1):5, 2024. [arXiv:2406.05046](#).
- [8] DESI Collaboration. DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations. *JCAP*, 02:021, 2025. [arXiv:2404.03002](#).
- [9] DESI Collaboration. DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints. *Phys. Rev. D*, 112(8):083515, 2025. [arXiv:2503.14738](#).
- [10] DESI Collaboration. Extended dark energy analysis using DESI DR2 BAO measurements. *Phys. Rev. D* 112(8), 083511, 2025. [arXiv:2503.14743](#).
- [11] E. Di Valentino, A. Melchiorri, O. Mena, S. Vagnozzi. Interacting dark energy in the early 2020s: a promising solution to the H_0 and cosmic shear tensions. *Phys. Dark Univ.*, 30:100666, 2020. [arXiv:1908.04281](#).
- [12] B. Feng, X. Wang, X. Zhang. Dark energy constraints from the cosmic age and supernova. *Phys. Lett. B*, 607: 35-41, 2005. [arXiv:astro-ph/0404224](#).

- [13] L. Heisenberg. Review on $f(Q)$ gravity. 2023. [arXiv:2309.15958](#).
- [14] A. Jiménez Cano. *Metric-Affine Gauge theories of gravity: Foundations and new insights*, Ph.D. thesis, 2022. [arXiv:2201.12847](#).
- [15] E. Komatsu et al. Five-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Cosmological interpretation. *Astrophys. J. Suppl.*, 180(2):330–376, Feb. 2009. [arXiv:0803.0547](#).
- [16] E. Komatsu et al. Seven-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Cosmological interpretation. *Astrophys. J. Suppl.*, 192, 2010. [arXiv:1001.4538](#).
- [17] M. Kunz, D. Sapone. Crossing the Phantom Divide. *Phys. Rev. D*, 74:123503, 2006. [arXiv:astro-ph/0609040](#).
- [18] M. Li, T. Qiu, Y. Cai, X. Zhang. On dark energy models of single scalar field. *JCAP*, 1212(04):003, 2012. [arXiv:1112.4255](#).
- [19] E. V. Linder. Exploring the expansion history of the universe. *Phys. Rev. Lett.* 90:091301, 2003. [arXiv:astro-ph/0208512](#).
- [20] E. V. Linder. Uplifting, depressing, and tilting dark energy, 2025. [arXiv:2506.02122](#).
- [21] J. E. Marsden, T. S. Ratiu. *Introduction to Mechanics and Symmetry*, Springer, 2nd Edition, 1998.
- [22] C. W. Misner, K. S. Thorne, J. A. Wheeler. *Gravitation*, W. H Freeman and Company, 1973.
- [23] S. Pan, W. Yang, E. Di Valentino, E. N. Saridakis, S. Chakraborty. Interacting scenarios with dynamical dark energy: observational constraints and alleviation of the H_0 tension. *Phys. Rev. D* 100(10):103520, 2019. [arXiv:1907.07540](#).
- [24] M. E. Peskin, D. V. Schröder. *An Introduction to Quantum Field Theory*, Perseus Books, 1995.
- [25] Planck Collaboration. Planck 2013 results. XVI. Cosmological parameters. *Astron. Astrophys.*, 571:A16, 2014. [arXiv:1303.5076](#).
- [26] Planck Collaboration. Planck 2015 results. XIV. Dark energy and modified gravity. *Astron. Astrophys.*, 594:A14, 2016. [arXiv:1502.01590](#).
- [27] T. Qiu, Y. Cai, X. Zhang. Null energy condition and dark energy models. *Mod. Phys. Lett. A*, 23(32):2787–2798, 2008. [arXiv:0710.0115](#).

- [28] X. Ren, T. H. T. Wong, Y.-F. Cai, E. N. Saridakis. Data-driven Reconstruction of the Late-time Cosmic Acceleration with $f(T)$ Gravity. *Phys. Dark Univ.* 32:100812, 2021. [arXiv:2103.01260](#).
- [29] D. Rubin et al. Union Through UNITY: Cosmology with 2,000 SNe Using a Unified Bayesian Framework. *Astrophys. J.*, 986(2):231, 2025. [arXiv:2311.12098](#).
- [30] D. N. Spergel et al. First year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Determination of cosmological parameters. *Astrophys. J. Suppl.*, 148:175–194, 2003. [arXiv:astro-ph/0302207](#).
- [31] D. N. Spergel et al. Three-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Implications for cosmology. *Astrophys. J. Suppl.*, 170(2):377–408, 2007. [arXiv:astro-ph/0603449](#).
- [32] J. A. Vázquez, D. Tamayo, G. García Arroyo, I. Gómez Vargas, I. Quirós, A. A. Sen. Coupled multiscalar field dark energy. *Phys. Rev. D*, 109(2):023511, 2024. [arXiv:2305.11396](#).
- [33] A. Vikman. Can dark energy evolve to the phantom? *Phys. Rev. D* 71(2):023515, 2005. [arXiv:astro-ph/0407107](#).
- [34] B. Wang, E. Abdalla, F. Atrio-Barandela, D. Pavón. Dark matter and dark energy interactions: theoretical challenges, cosmological implications and observational signatures. *Rept. Prog. Phys.*, 79(9):096901, 2016. [arXiv:1603.08299](#).
- [35] J.-Q. Wang, R.-G. Cai, Z.-K. Guo, S.-J. Wang. Resolving the Planck-DESI tension by nonminimally coupled quintessence. *Phys. Rev. D*, 113(8):083534, 2026. [arXiv:2508.01759](#)
- [36] B. Wang, Y. Gong, E. Abdalla. Transition of the dark energy equation of state in an interacting holographic dark energy model. *Phys. Lett. B*, 624(3-4):141-146, 2005. [arXiv:hep-th/0506069](#).
- [37] H. Wei, R. G. Cai, D. F. Zeng. Hesseence: A New View of Quintom Dark Energy. *Class. Quant. Grav.* 22:3189-3202, 2005. [arXiv:hep-th/0501160](#).
- [38] Y. Yang, X. Ren, Q. Wang, Z. Lu, D. Zhang, Y.-F. Cai, E. N. Saridakis. Quintom cosmology and modified gravity after DESI 2024. *Sci. Bull.*, 69:2698-2704, 2024. [arXiv:2404.19437](#).
- [39] Y. Yang, Q. Wang, C. Li, P. Yuan, X. Ren, E. N. Saridakis, Y.-F. Cai. Gaussian process reconstructions and model building of quintom dark energy from latest cosmological observations. *JCAP*, 2025(08):050, 2025. [arXiv:2501.18336](#).

- [40] Y. Yang, Q. Wang, X. Ren, E. N. Saridakis, Y.-F. Cai. Modified Gravity Realizations of Quintom Dark Energy after DESI DR2. *Astrophys. J.*, 988(1):123, 2025. [arXiv:2504.06784](#).
- [41] X.-F. Zhang, H. Li, Y.-S. Piao, X. Zhang. Two-field models of dark energy with equation of state across -1 . *Mod. Phys. Lett. A*, 21(03):231-241, 2006. [arXiv:astro-ph/0501652](#).